IDENTIFICATION OF ANTI-HEPATOCELLULAR CARCINOMA CANDIDATE DRUGS FORDRUG REPOSITIONING USING GENE SIGNATURES

THESIS

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By

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CERTIFICATE

This is to certify that the dissertation thesis entitled "***Identification of anti-hepatocellular carcinoma drug candidates for drug repositioning using gene signatures***" submitted by **ISHAN SORATHIYA** (PRN NO. 2020033800123995) as part fulfillment of the requirements for the award of the degree of **Bachelor of Science (Hons) in Zoology**, embodies the work carried out in the **Department of Zoology, Faculty of Science**, **The Maharaja Sayajirao University of Baroda**, Vadodara, Gujarat. The content presented in this dissertation thesis incorporates the original findings of independent research work carried out by the candidate himself. The content of this thesis has not been submitted elsewhere for the award of any degree.

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DECLARATION

I hereby declare that the thesis entitled "***<u>Identification of anti-HCC candidate drugs for drug repositioning using gene signature</u>***" submitted in partial fulfilment for the award of **the Degree of Bachelors of Zoology (Hons)** is original and has been carried out at The Maharaja Sayajirao University of Baroda, in accordance with the results obtained by me. The information derived from the literature has been duly acknowledged in the text and a list of references provided. No part of this dissertation was previously presented for another degree or diploma at this or any other institution. I hereby confirm the originality of the work and that there is no plagiarism in any part of the dissertation.

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Content

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Abbreviations

ITA.LI.CA	Italian Liver Cancer
ABL2	Abelson murine leukaemia viral oncogenehomolog 1
AFP	Alpha feto-protein
AKT	Ak strain transforming
ALD	alcoholic liver disease
AML	acute myeloid leukaemia
ASR	age-standardized rate
ATP7B	ATPase activity
AURKA	Aurora kinase A
BCL2	B-cellleukaemia/lymphoma 2 protein.
BCLC	Barcelona Clinic Liver Cancer
BDE-HCC	Bile duct epithelium like HCC
BIRC5	Baculoviral IAP Repeat Containing 5
Bub1	Budding uninhibited by benzimidazoles 1
CCLE	Cancer cell line encyclopedia
CCNA2	Cyclin A2
CCNB1	Cyclin b1
CDH1	Cadherin
CDK4	Cyclin dependent kinase
cDNA	Complementary DNA
CHEK1	Checkpoint kinase 1
CLIP	Cancer of the Liver Italian Program
COSMIC	Catalogue Of Somatic Mutations In Cancer
DCP	Desaturation-γ-lock-up-thrombin

DDR1	Discoidin Domain Receptor Tyrosine Kinase 1
DEGs	Differentially expressed genes
EGFR	Epidermal growth factor receptor
EMA	European Medicines Agency
EMT	Epithelial-mesenchymal transition
EpCAM	Epithelial calcium adhesion molecule
FAH	Fumarylacetoacetate Hydrolase
FDA	Food and Drug administration
GEO	Gene expression omnibus
GLOBOCAN	Global Cancer Incidence, Mortality and Prevalence
GO	Gene ontology
GPC-3	Glypican 3
HB	Hepatoblast
HBV	Hepatitis B-virus
HC	Hepatocyte
HCA	Hepatocellular adenoma
HCC	Hepatocellular carcinoma
HCV	Hepatitis C-virus
HDAC2	Histone Deacetylase 2
HFE1	Hemochromatosis
HKLC	Hong Kong Liver Cancer
HMBS	Hydroxymethylbilane synthase
HP-HCC	Hepatocyte progenitor like HCC
HpSC-HCC	Hepatic stem cell like HCC
HSC	Hepatic-stellate cells
IARC	International Agency for Research on Cancer
ICC	Intrahepatic cholangiocarcinoma

ID2	Inhibitor of DNA binding
IL6	Interleukin 6
IP	Intellectual patent
KEGG	kyotoencyclopaedia of genes and genomes
MAD2	Mitotic arrest deficient 2
MCM3-7	Minichromosome maintenance complex component
MDR	Multi-drug resistance
MESIAH	Model to Estimate Survival In Ambulatory HCC patients
MH-HCC	Mature hepatocyte like HCC
MHRA	Medicines and Healthcare products Regulatory Agency
MMP9	Matrix metalloproteinase 9
NAFLD	Non alcoholic fatty liver disease
NASH	Non-alcoholic steatosis hepatitis
NCBI	National Center for Biotechnology Information
NCI	National cancer institute
OPN	Osteopontin
OS	Overall-survival
PDGFR-β	Platelet-derived growth factor receptor beta
PDK1	3-phosphoinositide-dependent protein kinase-1
PI3K	Phosphatidylinositol 3-kinase
PIK3CA	Phosphatidylinositol-4,5-Bisphosphate 3-Kinase Catalytic Subunit Alp
PNCA	Nicotinamidase/pyrazinamidase
QRT-PCR	Quantitative Reverse Transcription Polymerase Chain Reaction
RAS	Rat sarcoma
RNAi	Inhibitor RNA
ROS	Reactive oxygen species
ScRNA	Small cytoplasmic RNA

SERPINA1	Serine protease inhibitor (serpin)
SGK	Serine/threonine-protein kinase
SRC	Proto-oncogene tyrosine-protein kinase Src
SVM	Support vector machine
TCF/LEF	T-cell factor/lymphoid enhancer factor
TCGA	The Cancer Genome Atlas
TKI	Tyrosine kinase inhibitor
TNFAIP3	TNF Alpha Induced Protein 3
TOP2A	Topoisomerase II alpha
TP53	Tumor protein 53
TTK	Threonine tyrosine kinase
TYMS	Thymidylate synthase
UROD	Uroporphyrinogen decarboxylase
USFDA	U.S. Food and Drug Administration
VEGFR	Vascular endothelial growth factor
VI	Vascular invasion
Wnt	Wingless

Chapter 1

Introduction

1.0 Introduction

Hepatocellular carcinoma (HCC) is one of the common primary malignant solid tumors of hepatocytes [1,2]. The accumulation of several genetic alterations and epigenomic changes of tumor that differ in risk factors is a defining feature of HCC occurrence [3,4]. It is a highly heterogeneous disease both at the pathological and molecular levels, stratified into intra and inter-tumor heterogeneity which limits the diagnosis and treatment of HCC patients as it is asymptomatic in early stages [5,6]. Several factors can cause the onset of HCC such as HBV & HCV, Aflatoxin, heavy alcohol consumption, obesity, type-2 diabetes, and nicotine intake. HCC is also known to occur in people with chronic liver diseases, such as cirrhosis caused by hepatitis B or hepatitis C infection [2,7]. Chronic fibrotic liver disease caused by viral or metabolic aetiologies is a high-risk condition for developing hepatocellular carcinoma (HCC) [8].

Hepatocellular carcinoma (HCC) is responsible for a great majority of liver cancer diagnoses as well as deaths [9], making it one of the most common types of cancer [9]. A recent GLOBOCAN (Global Cancer Observatory) report reveals that liver cancer is the sixth most prevalent cancer [10]. In addition, HCC has also been found to be the third predominant cause of cancer related deaths worldwide [6].

A ‘gene signature’ can be defined as a change in expression in a group of geneswith better specificity for diagnosis, prognosis, or therapeutic response [11]. So, gene expression signature can be broadly divided into (i) Prognostic gene signature (ii) Diagnostic gene signature and (iii) Predictive gene signature [12-15]. Additionally, recurrence gene signature, epithelial mesenchymal transition gene signature, metastatic gene signature and survival gene signatures have also been generated to predict different outcomes of HCC [1,3,5,16-24]. Moreover, according to these gene signatures, cancers at the same stage may have various genetic blueprints, this may increase the sensitivity and specificity of HCC identification [25].

Of all the global profiling technologies used as of now, DNA microarray and associated RNA-sequencing techniques are some of the most popular techniques for obtaining molecular signatures for the prediction of HCC recurrence. By hybridizing cDNA to DNA probes on a chip, DNA microarrays enable high-throughput analysis of these data sets [26]. RNA-sequencing is frequently carried out at a high throughput scale, like the DNA microarray, therefore it should be able to use data from one approach to confirm the other [27]. Particularly for the purpose of transcriptional cell type classification in various malignancies, scRNA-seq has become a valued technique, as a result of its ability to identify a rare cell population and differentiate between different cell types in a heterogeneous population [28]. For liver cancer, more than 300 microarray studies have been published identifying genes dysregulated in HCC [29].

It has been observed that gene signatures not only help to predict diagnosis, prognosis, or overall survival but alteration in the gene expression of the signature member gene using inhibitors or RNAi, could improve the diagnosis, prognosis, or overall survival of HCC patients. Inhibitor or RNAi based gene expression manipulation can also reverse the signature. Schiffer identified that prognostic liver signature member EGF has an important role in development of HCC and inhibition of EGF mediated signaling using epidermal growth factor receptor (EGFR) inhibitor Gefitinib exerted antitumoral effect on the development of HCC [8,30]. In another study, Sorafenib, a multiple-target tyrosine kinase inhibitor (TKI) was found to be capable of facilitating apoptosis, mitigating angiogenesis and suppressing tumor cell proliferation effects and extended total median survival in advanced HCC patients by inhibiting prognostic liver signature (Ouyang et. Al, 2020) gene platelet-derived growth factor receptor (PDGFR-β) and EMT signature (Kim et. al, 2010) gene vascular endothelial growth factor receptor (VEGFR)-[14,20,21]. In addition, Tongqi et. al also demonstrated that targeting BCL2 from the fibrosis progression signature (daughter signature of prognostic liver signature by Hoshida et. al) can reverse the activation of hepatic stellate cells, which are a primary cause of liver fibrosis progression, an intermediated condition of HCC progression [31].

Inhibition of a metastatic signature gene (Rossler et. al, 2010) osteopontin (OPN), using neutralizing antibodies, short peptides, or lentivirus-mediated RNA interference can prevent HCC cell invasion in vitro and decreases lung metastasis in mice, suggesting that OPN may be a promising therapeutic target for metastatic HCC [23,32-34].

The process for the approval of a new drug is expensive and can take 10-15 years [35]. Finding novel therapeutic uses for currently available medications is referred to as drug repositioning. Investigating medications that were once approved for one application but may perhaps be beneficial for other medical ailments can be part of this. Investigating the efficacy of approved or discarded drugs for new indications using an approach called drug repositioning/purposing, can in fact overcome some of the obstacles in drug discovery, such as the necessity to meet quality standards [36-38]. With the aid of drug repurposing, there is a greater chance of success and at least known and approved negative effects [39]. The ultimate aim of repurposing research is to reduce the time and expense of medication discovery and development by properly anticipating clinical utility and using predictions to enhance health and quality of life. Drug repurposing technologies are conducted to automate and streamline clinical trial procedures [37,40]. There are several legal intellectual property (IP) and patent rights which hold a barrier to drug repurposing [41]. Any changes to the drug's structural makeup are not included in the idea of drug repositioning. Repositioning, on the other hand, uses a new indication of either the biological features for which the medicine has already received approval or the side effects of a drug that cause them [42]. In diseases, drugs can be analyzed for phenotypic similarity, regardless of their therapeutic indication. If two drugs indicated in different diseases have a high similarity score, they may be effective in both indications [43].

As discussed above hepatocellular carcinoma (HCC) is a type of primary malignant solid tumor of the liver. A defining aspect of HCC incidence is the aggregation of multiple genetic mutations and epigenetic modifications of cancer that differ in risk variables. It is a pathologically and molecularly heterogeneous illness, classified into intra and inter-tumor heterogeneity, which limits the diagnosis and treatment of HCC patients because it is silent in the early stages, so it is essential to find a therapeutic strategy to prevent HCC mortality and incidence worldwide. Here we used the concept of drug repositioning for hepatocellular carcinoma by identifying altered and/or unaltered genes from the gene signatures. The alteration in genes from the gene signature provides a basis for type of malignancy shown by HCC and with the help of gene targeting drugs we might inhibit these differentially expressed genes in the gene signatures of different aspects of HCC. So, we hypothesis that

(i) High differential expression of gene signature members could provide a basis for finding a targetable gene candidate

(ii) Clinical status of candidate gene targeting drugs/compounds could be employed for devising therapeutic strategy against HCC

Since genes from signatures have been proved to be therapeutically targetable, in the current study, we collected genes from multiple gene signatures involving HCC. Those genes were further filtered on the basis of their fold change expression in HCC cases with the help of OncoDB database which employs TCGA gene expression data in various cancers. Here we specifically extracted the gene expression values from mRNA sequencing data of hepatocellular carcinoma tumors compared with normal liver tissue (LIHC, n= 371 and normal, n= 50). Next, we selected the upregulated genes with ≥50% DEG because we are specifically interested in inhibitors mediated suppression of upregulated genes. After selecting genes, we extracted the inhibitors which are able to suppress the effect of upregulated genes on the overall gene expression. Later, clinical trials were retrieved in which those inhibitors were specifically used against various cancers. On the basis of clinical trials' results, we screened several drugs which we propose for drug repositioning against HCC.

This whole study was divided into below mentioned objectives -

1. Identification of gene signatures and selection of DEGs of gene signature members.
2. Extraction of selected DEGs inhibitor molecules from CLUE database.
3. Selection of inhibitor molecules on the basis of current cancer clinical trials.

Chapter 2

Review of Literature

2.0 Review of literature

2.1 Hepatocellular carcinoma

There are mainly two types of liver cancer (i) HCC and (ii) Intrahepatic cholangiocarcinoma (ICC) while other types are hemangiosarcoma & hepatoblastoma. [2]. HCC arises from hepatocytes (main parenchymal cells of the liver), while Intrahepatic cholangiocarcinoma (ICC) originates from bile ducts [2]. Primary liver cancer includes HCC comprising 90% of cases [44]. The pathophysiology of HCC is a complex multistep process 33479224. Cirrhosis is the strongest risk factor for HCC [45,46]in which hepatocytes gradually accumulate a wide range of genetic mutations and epigenetic alterations. During this process, several risk factors that cause DNA mutations are linked to particular mutational signatures [47,48]. Hepatocellular carcinoma (HCC) arises from hepatocytes and accounts for 90% of primary liver cancer. Additionally, secondary liver cancer occurs when a tumor from another part of the body metastasizes to the liver, which usually progresses into HCC. Most secondary liver cancers originate from colorectal cancer. Approximately 70% of colorectal cancer patients develop secondary liver cancer [2]. The occurrence of HCC is commonly observed in males to a greater extent as compared to females [49].

In addition to cirrhosis induced HCC, specific viral processes may aid in the development of hepatocarcinogenesis, prolonged necroinflammation and the production of reactive oxygen species can result in chromosomal alterations and finally the malignant transformation of proliferating hepatocytes which shows HCC is a complicated multistep pathogenesis. [50]. Early stages of development of HCC are usually found to be asymptomatic. Symptoms are diagnosed in advanced or late-stage disease [2].

2.2 Risk factors

Major risk factors for HCC include chronic alcohol consumption, infection of hepatitis B and hepatitis C virus, and non-alcoholic fatty liver disease [51]. HBV is a DNA virus that can integrate into the host genome inducing insertional mutagenesis, leading to oncogene activation which concludes HBV increases the risk of HCC even in the absence of cirrhosis whereas HCV is an RNA virus that does not integrate into the host genome and the risk of HCC is primarily limited to those who develop cirrhosis [46] (Figure1,2). Low levels of alcohol use had no association with HCC risk, whereas excessive alcohol use (12-24 grams/day) was linked to a 16% increased risk of ALD 24631946 and heavy consumption of alcohol (>40 grams/day) has been found to be associated with an almost four-fold increased risk of ALD among women, but only a 59% increased risk among men. In addition, alcohol may have a stronger association with HCC risk among women than men [52]. Another factor obesity has continuously been associated to an increased risk of developing alcohol-related fibrosis and cirrhosis, presumably due to a synergistic interplay between alcohol and steatosis-induced lipotoxicity[53-55].

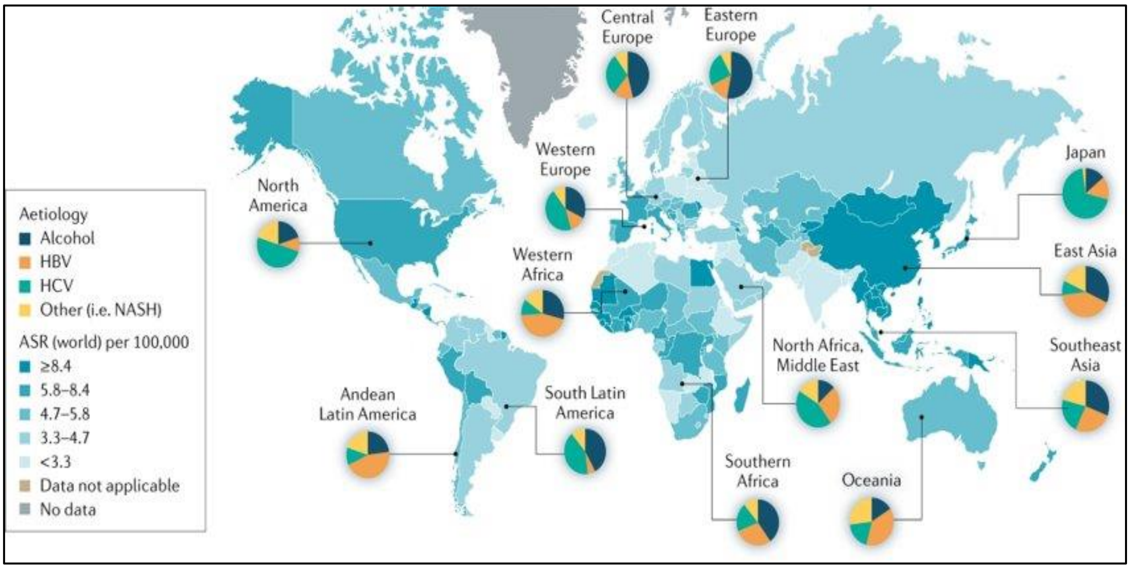


Figure 1: Various HCC associated aetiologies, (http://gco.iarc.fr/today)

Mendelian genetic predisposition to hepatocellular carcinoma is a risk factor associated aetiology. HCC predispositions are seen in a variety of genetic and physiological disorders, primarily by cirrhosis development. These unusual iron-related disorders include hereditary hemochromatosis (HFE1 gene) [56], excessive accumulation of copper in liver (Wilson disease, ATP7B gene) [57], Type 1 tyrosinemia (FAH gene), acute intermittent porphyria (HMBS gene), Cutanea tarda (UROD gene) and the alpha-1-antitrypsin deficiency (SERPINA1 gene) [58,59].

The etiologically key determination for HCC varies by country to country for example, in Mongolia, HBV & HCV and co-infections of HBV carries with HCV or hepatitis delta viruses, as well as alcohol consumption are responsible for higher rates [44]. Cofactors that also increase risk among HBV carriers include demographic characteristics (e.g., male sex, older age, Asian or African ancestry, family history of HCC), viral factors (e.g., high HBV replication levels, HBV genotype, infection duration, coinfection with HCV or HIV), and environmental exposures (e.g., aflatoxin, alcohol, tobacco, obesity, diabetes) [60]. Viral hepatitis & alcohol remain important etiological risk factors while NAFLD is a major dominating cause of HCC. The global incidence of viral hepatitis related malignancies has declined since the 2000s because of the implementation of neonatal HBV vaccination programs& the availability of highly effective antiviral treatments for HBV & HCV [61]. Multiple risk factors can cause liver cancer which may vary substantially by geographical region [44]. These factors include a genetic predisposition, reciprocal interactions between viral and non-viral risk factors, the cellular microenvironment and various immune cells, and the severity of the underlying chronic liver disease [45]. The relevance of non-viral risk factors is becoming more important on the burden of liver cancer. Elimination of hepatitis remains the key strategy for primary prevention of liver cancer globally, as HBV & HCV infections account for 56% and 20% of liver cancer deaths worldwide, respectively [44]. Unsafe injections and other invasive procedures cause high HCV prevalence and may lead to liver cancer [44]. The incidence of HCC varies by geographical region and ethnicity, which is largely attributed to the prevalence of major risk factors like NAFLD, NASH, HCV and HBV and rarely in haemochromatosis or hereditary tyrosinemia type 1. Exposure to aflatoxin B1 is especially relevant in Asia, where it overlaps with HBV infection [62]. Due to the complicated aetiology of liver cancer and the different molecular subtypes observed in individuals, early diagnosis is difficult [63]. Although the prevalence of virally driven hepatocellular carcinoma (HCC) has declined in recent years while the incidence of NAFLD and NASH related liver cancer has increased [7]. NAFLD is part of a multisystem disease and is considered the hepatic manifestation of the metabolic syndrome [64]. The International Agency for Research on Cancer (IARC) listed aflatoxin B1 as a class I carcinogen in 1987 [10].

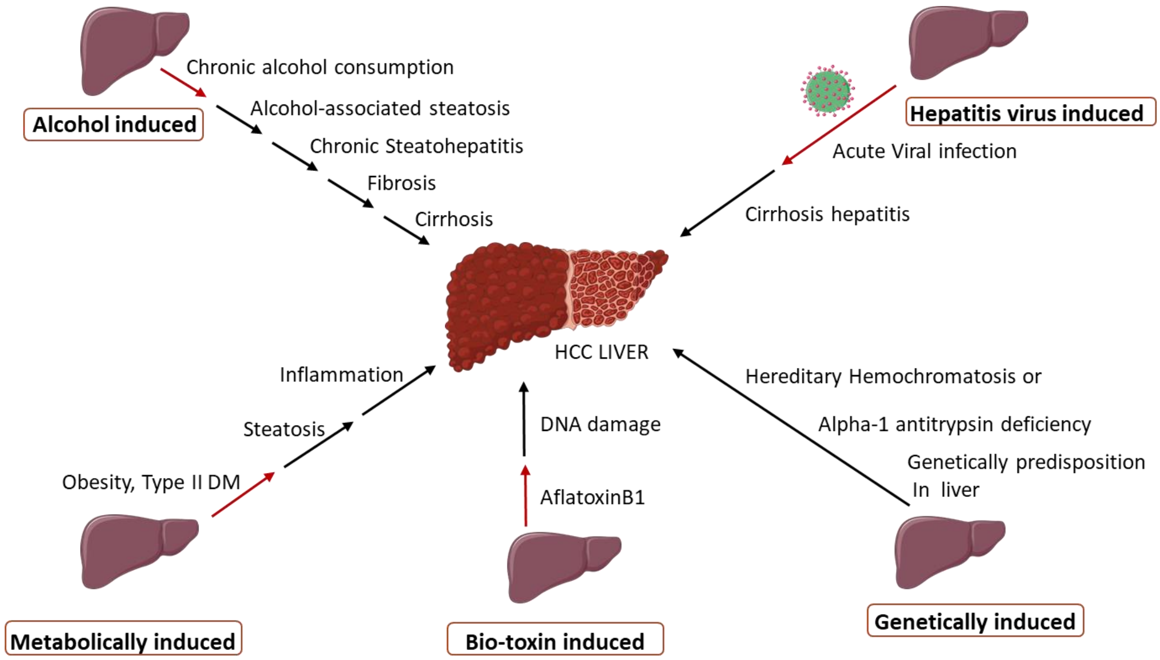


Figure 2: Major HCC causing risk factors and the HCC development associated stages.

2.3 Epidemiology and prevalence

Primary liver cancer is the sixth most diagnosed cancer and third leading cause of cancer death worldwide in 2020, with approximately 906,000 new cases & 8,30,000 deaths accounting for more than 80% of primary liver cancer incidence worldwide [2,44]. Rates of incidence & mortality are observed 2-3 times higher in males than females in most geographical regions [44]. Liver cancer is the sixth most common malignancy, according to a recent GLOBOCAN (World Cancer Observatory) survey in 2020, following gastric cancer, lung cancer, thyroid cancer, and colorectal cancer [10].

The highest incidence of hepatocellular carcinoma (HCC) is observed in East Asia, with Mongolia demonstrating the highest incidence of HCC worldwide (Figure 3). Hepatitis B virus (HBV) is the major aetiological factor in most parts of Asia (except Japan), South America and Africa; Hepatitis C virus (HCV) is the predominant causative factor in Western Europe, North America and Japan, and alcohol intake is the aetiological factor in Central and Eastern Europe. Non-alcoholic steatohepatitis (NASH), the main aetiology included in the category 'Other', is a rapidly increasing risk factor that is expected to become the predominant cause of HCC in high income regions soon as oer age-standardized incidence rate [45]. European cohort-based studies have reported that, among persons who died of a liver-related cause, HCC was the responsible cause in 54% 15057740 to 70% [65]of patients with compensated cirrhosis of different etiologies and in 50% of patients with HCV-related cirrhosis [66]. Cirrhosis of all etiologies may be complicated by HCC, but persistent HBV or HCV infection account for over 80% of HCC cases worldwide [67]. The distribution of liver cancer presented significant geographic disparities, and China has contributed to nearly half of the global cases alone [51]. Liver cancer is one of the leading causes of having a high mortality rate in Mongolia [44].

The prevalence of cirrhosis in patients with HCC is about 80%–90% in autopsied series worldwide, and, therefore, approximately 10% to 20% of cases of HCC develop in a patient without cirrhosis [68]. Among HCC cases with cirrhosis, HCV infection was identified in 27%–73%, HBV infection in 12%–55%, heavy alcohol intake in 4%–38%, and hemochromatosis and other causes in 2%–6%, leaving 4%–6% of the total number of cases without an identified cause [50]. Cirrhosis is a significant co-factor in the development of liver cancer. The annual incidence of liver cancer was estimated at 2%−4% among individuals diagnosed with cirrhosis.30970190. The incidence and mortality of HCC continue to rise especially because of the obesity pandemic which gives rise to NAFLD, and globally HCC mortality is expected to increase another 41% by 2040 [44].

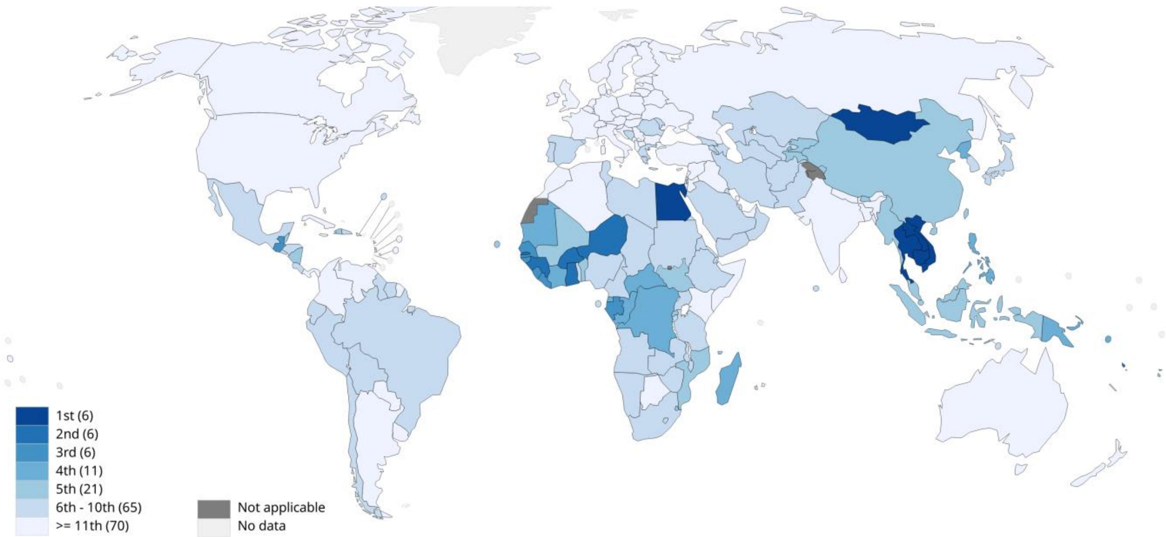


Figure 3: Ranking pf countries on the basis of global and countrywise HCC incidence rate (http://gco.iarc.fr/today)

The global HCC incidence ratio of men to women is 2.8:1[2]. In most countries, incidence rates among men are 2 to 4-fold higher than rates among women 32319693[9]. Men are at higher risk of developing liver cancer [2]. For men, the prevalence of liver cancer increased from 268.2 thousand in 2011 to 310.6 thousand in 2020 and so had the age-standardized prevalence proportion from 38.8 per 100,000 in 2011 to 41.9 per 100,000 in 2020. While the liver cancer prevalence proportion is found lower for women compared to men, with the age-standardized prevalence proportion from 14.6 per 100,000 to 15.9 per 100,000 between 2011 and 2020 [10]. The prevalence of HCC has been linked to population factors, especially in cirrhotic patients. Age is another significant risk factor, with people over 70 years old reporting the highest age-specific incidence [69]. In China, HCC ranks first both in incidence and mortality in males under 60 years old [70]. Age- and sex-standardized mortality is the lowest in Japan, and highest in Mongolia [71]. As a result of the high prevalence of endemic HBV infection in East Asia, the risk of HCC in men (40 years of age) and women (50 years of age) is more than the cost-effectiveness thresholds, which justifies surveillance programs. The diagnosis of hepatocellular carcinoma peaks in people aged between 60 to 70 years and predominantly affects men indicating, ageing is a strong risk factor, with the highest age-sex specific incidence[69,72]. In most populations, incidence rates of HCC and age are directly correlated until approximately 75 years of age. The median age at diagnosis, however, is generally somewhat younger. In the US, for example, the median age at diagnosis among men is between ages 60 and 64 years, while the median age among women is 65–69 years. By contrast, in Africa, there is a significant difference in median age at diagnosis between Egypt (58 years) and other African countries (46 years) [9]. The prevalence and age at which key risk factors are acquired are largely, but not totally, responsible for the significant variation in HCC incidence by geographic region, age, sex, and race/ethnicity. 32319693[9].

The average 5-year survival rate of HCC patients in the US is 19.6% but can be as low as 2.5% for advanced, metastatic disease [2].

Diagnosis, Prognosis, survival rate and recurrence of HCC patients:

Besides prevention, early tumor detection is one of the most important methods for improving the prognosis of HCC which can be achieved by imaging modalities which are an important tool in the monitoring and diagnosis of HCC [73]and the prognosis of the disease depends on the tumoral stage 21909721. The main HCC screening techniques have been abdominal ultrasonography and serum AFP [74].

Due to the excellent sensitivity of dynamic imaging (computed tomography and MRI) in identifying HCC more than or equivalent to 2 cm, a biopsy is not required to confirm the diagnosis of HCC[75]because patients with cirrhosis and coagulopathy are more likely to experience bleeding problems after a biopsy28834467 Moreover, tumor seeding following biopsy has been estimated to be as high as 2.7% [76].

The molecular markers commonly used for the diagnosis of HCC are alpha-fetoprotein (AFP) [77], Desaturation-γ-lock-up-thrombin (DCP) [78]and phosphatidylinositol proteoglycan-3 (GPC-3) [50]. AFP is the oldest biomarker still in use for HCC detection and prognostication as it increases by 60% in HCC [79,80]. Abelev et al.'s 1963 study on mice and Tatarinov's 1963 and 1964 studies on liver cancer patients were the first to show that AFP is a marker for liver tumors[81-85]. Up to 70% of HCC patients show higher serum levels of AFP, which highlights the importance of the AFP level therefore AFP is used as a clinical diagnostic and prognostic tumor marker for HCC [84]. It is widely accepted that serum Alpha-fetoprotein(AFP) levels are significantly related to survival also and exerts inverse correlation with survival. Lee et. al (2004), segregated HCC cohorts into Cluster A and Cluster B based on their AFP level serum. Cluster A showed a higher level of AFP (62.5%) while cluster B showed lower level of AFP2010(42%). The overall survival of Cluster B was higher than Cluster A as the Kaplan-Meier survival curve of cluster A indicated poorer survival. Expression of cell proliferation markers such as PNCA was high in cluster A and showed higher expression of anti-apoptotic genes [17]. Yamashita et. al reported that based on EpCAM-coexpression and AFP levels classification to be associated with specific HCC subtypes that resemble hepatic cell maturation lineages, indicating a potential cellular origin of HCC and activated molecular pathways which may provide new insights into potential therapeutic targets for HCC treatment [1].

There are many HCC staging systems worldwide like Barcelona Clinic Liver Cancer (BCLC), HCC-Okuda, CLIP (Cancer of the Liver Italian Program) score, MESIAH (Model to Estimate Survival In Ambulatory HCC patients) score, ITA.LI.CA (Italian Liver Cancer) score, HKLC (Hong Kong Liver Cancer) staging, and the Alberta algorithm. Despite the fact that several regions of the world have established new staging systems for HCC, there isn't a staging system that allows for a comparison of existing management strategies among diverse populations [86]**.** Based on strong clinical evidence, the Barcelona Clinic Liver Cancer (BCLC) staging system remains the most proven and dependable prognostication approach [86]which connects HCC prognosis with the optimal treatment strategy so to improve the clinical outcomes and for a proper risk stratification. Stephanie Roessler et. al (2010) used gene classifiers to improve the BCLC staging and found that the metastasis risk classifier improved survival prediction when combined with BCLC specially for those in the early stage of HCC [23,87].

According to the classic tumor evolution concept, a primary tumor is initially benign and then gains mutations that allow a few tumor cells to spread ([88,89]). Long-term survival should be improved if a tumor is found and treated before it spreads. As a result, early identification is critical for improving patient outcomes. Recent research suggests that even if tumors are detected early, they may have already completed most of the processes on their road to metastasis [90,91]. The genetic machinery that promotes metastasis, for instance, may be constructed into the tumor from the start, according to genomic analysis of primary colon tumors and matched metastases [92].

If patients with high risk for recurrence could be identified in advance, patients with low risk of recurrence would not undergo unnecessary financial strain and side effects [23]. Intrahepatic recurrence, which occurs in 30–50% of patients who receive hepatic resection, is a significant barrier to its treatment. Therefore, the possibility of surgery as a treatment for hepatocellular carcinoma is constrained by intrahepatic recurrence [20,93,94]. However, Suk Mei Wang et al. (2007) reported recurrence of HCC was predominant in males, with its ratio to female of 4.3:1. HCC recurrence is a serious complication that occurs in 50% of cases three years after removal of the initial tumor[95]. Kurokawa et al. studied gene expression in 100 HCC patients utilising a quantitative reverse transcriptase polymerase chain reaction (qRT-PCR)-based array platform containing 3072 genes [96]. The researchers selected 92 genes with different expression patterns that differed significantly between recurrence and recurrence-free patients. Two different recurrence types in HCC are recognised. Early recurrence develops from primary cancer cells spreading to the surrounding liver and is typically noticed during the first two years following surgery on the other hand late recurrence, which is often noticed after more than two years after surgery, seems to be caused by chronic liver damage known as the "field effect" and results in de novo tumors that are independent of the main tumors that were surgically removed [97,98]. To identify important genes that may indicate field effects in the liver that eventually result in the formation of HCC. Based on such observations Hoshida et al. characterized gene expression data from nontumor surrounding tissues from HCC patients [99].

Vascular invasion, level of tumor differentiation (size and number), AFP levels and multi- nodularity are the main histological characteristics that predict HCC recurrence and prognosis. After the removal of a single tumor, overall survival rates at five years are greater than 50%, although rates of 20% to 30% have been reported for three or more nodules [95,100-102]. Even in individuals with a single tumor less than 2 cm, the recurrence of HCC following hepatic resection continues to be extremely difficult. Recurrence rates can reach 70% at 5 years and from which intrahepatic recurrences account for 80% to 95% of postoperative cases [103,104].

2.4 HCC gene signatures

In the late 1990s, the development of whole-genome expression profiling had a significant impact on biomedical research which reported that tumors varied greatly at the molecular level even though they are at the same clinical stage. Identification of particular expression profiles, or more specifically, by a signature, for Acute Myeloid Leukemia (AML) was first reported in 1999 [105] Classification of cancer is divided into two challenges in this study: class discovery and class prediction.

The invention of microarray technologies, which allow us to evaluate numerous genes simultaneously, has resulted in a new age in medical science 8944026. An extensive amount of knowledge on the genes that are differentially expressed in different types of cancer is still being provided through microarray studies of human tumor samples and bioinformatics meta-analyses [106-108]. Large-scale transcript analysis in many samples has been made possible by array-based technologies, which laid the groundwork for cancer classification based on expression patterns. For example, the complementary DNA microarray investigation of cancer gene expression has provided valuable insights into their characteristics like prognosis [109]. DNA microarray and related sequencing techniques are the most used method to obtain molecular "signatures" for prediction of HCC recurrence [9].

Over the past 15 years, several research focusing on the gene expression profile of HCC have been published. HCC's molecular profiling poses difficulties that are uncommon in other human tumors because of heterogeneity of HCC [110,111]. Two general approaches have been used for the analysis of cancer genomic data to identify molecular subgroups that are significantly correlated with clinical outcomes. (I) Supervised technique is to identify a set of factors, such as the frequency of tumor mutations or expressed genes, that can be used to predict clinical outcomes in patients, such as survival, recurrence, therapy response, or any class of interest whereas, (II) Unsupervised technique is to identify an entirely new subset (or cluster) of patients that have never been identified before or to identify similarities among a group of patients that were previously thought to be clinically distinct patients [112]. To more accurately pridict various outcomes of HCC, gene signatures have been combined with clinical biomarkers such as PCNA. A "gene expression signature" is a unique characteristic pattern of single or combined group of gene expression resulting from an altered or unaltered biological process that has been shown to be particular in terms of diagnosis, prognosis, or therapy response prediction [11]. The concept of diagnostic and prognostic gene signatures has been introduced based on the cancer classification/subclass concept [113].

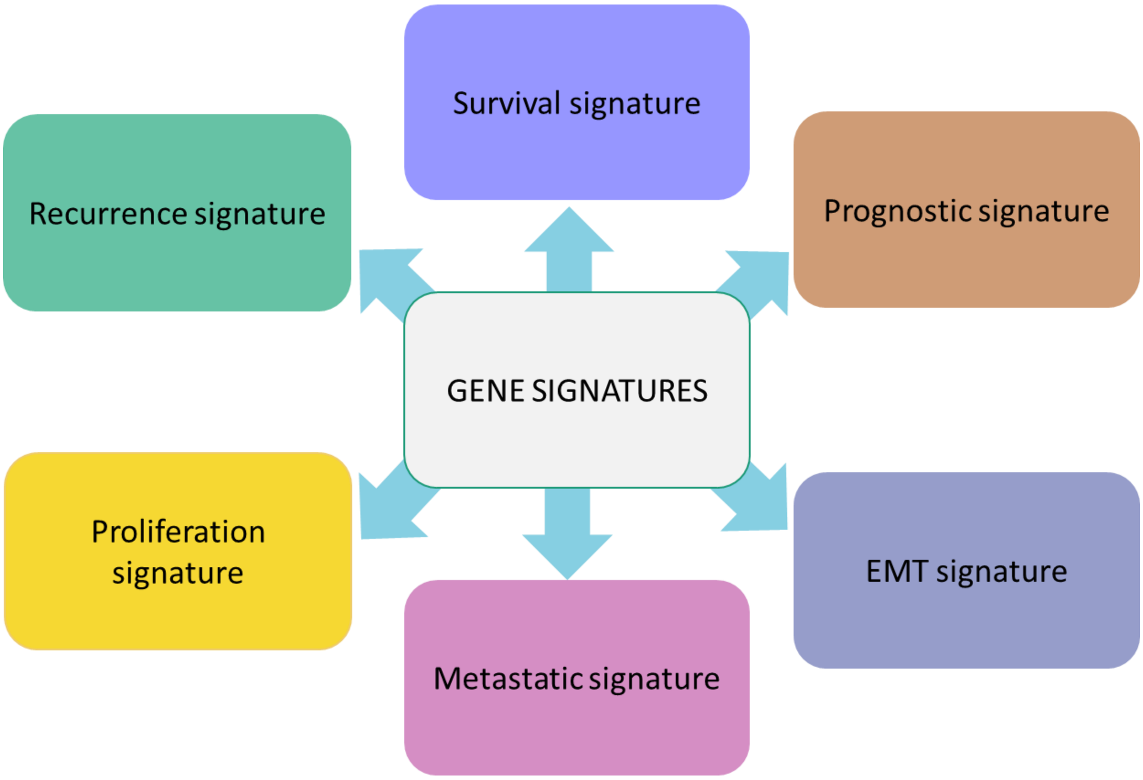


Figure 4: Different types of HCC gene signatures

A prognostic factor is an objectively measurable clinical or biologic feature that provides information on the likely outcome of a malignant disease in a certain clinical circumstance (treated or not treated). These prognostic indicators are useful for locating cancer patients who face a high probability of relapse or passing away. Contrarily, a predictive factor is a clinical or biological characteristic that indicates the likely benefit from treatment, independent of prognosis. These predictors can be used to pinpoint patient subpopulations that will most likely benefit from a particular medication [11,114].

Most prognostic signatures, including the HCC subtyping proposed by Boyault et al. [3], [115], were created utilising transcriptome data. Similarly, epithelial cell adhesion molecule and AFP based signature [1], survival-based signature including two clusters A and B [17] and hepatoblast-like [116] like gene signatures were generated for prognosis.

According to the prognosis, HCC patients can be categorised according to signatures extracted from nearby cirrhotic tissue, according to several studies. These classes presumably record oncogenic signals corresponding to the so-called "field effect."

Using a 17400-gene cDNA microarray, Chen et al. (2002) examined 102 primary HCC tumor samples and 74 non-tumor samples for DEGs. They showed increased expression of genes involved in DNA replication, such as MCM3-7, TYMS, and PCNA, as well as genes implicated in G2/M progression, such as CDC2, CDC20, PLK1 and mitotic regulators like MAD2 and Bub1. Reduced expression of genes unique to the liver is a sign of dedifferentiation or loss of liver function in hepatocytes. In addition to above mentioned genes, p53 is considered as a main component of this proliferative gene signature [16].

In 2004, Thorgeirsson's team used microarray analysis to examine the gene expression profile of 91 primary HCC tumors in humans. Additionally, they determined that the proliferative cluster of genes' increased expression was the strongest indicator of a poor result. This proliferative gene profiles successfully identified Clusters A and B, two groups of HCC patients with significantly different median survival times of 30 months and 90 months, respectively. PCNA, Bub3, MCM2, 6, and 7 were all part of the proliferative gene cluster, as were the cell cycle regulators cyclinA2 (CCNA2), cyclinB1 (CCNB1), CDC2 associated protein 2 (CKS2), and cyclin-dependent kinase 4 (CDK4) [17].

Two kinds of HCC were discovered by Thorgeirsson et al. (2006) using global gene expression studies on 139 human HCCs. Hepatoblast characteristics were seen in one subtype (HB), while differentiated hepatocyte characteristics were present in the other (HC). The approach used in this work includes comparing the gene expression profiles of three different species (human, rat, and mouse) [116]. Alpha-fetoprotein (AFP) was expressed at comparable quantities in both HB and HC subtypes. The HB subtype co-clustered with the proliferative set of genes that characterised the poor survival Cluster A of human HCCs, according to further examination of the gene expression profiles using hierarchical cluster analyses. The HB subtype had the lowest survival rate among the HB and HC subtype in Cluster A, where the HC subtype was divided into Clusters A (proliferation signature-positive) and B (proliferation signature-negative) [17]. Similarly, two primary clusters of HCC tumors were found by Boyault et al. (2007) and further divided them into six different subgroups (G1–G6). These G1-G6 subgroups were further distinguished by their relationship with distinctive clinical and molecular/genetic changes, including as viral infection, activation of the PI3K/AKT pathway, and p53 mutations, offering additional descriptors for accurate classification of HCCs and chromosome instability was more common in subgroups G1–G3 and was linked to a poor prognosis. [3]. In addition to above mentioned study, it has been observed that multiple human malignancies contain a proliferative cluster, pointing to a shared mechanism in tumor growth [117].

A 186-gene signature enriched in gene sets linked to inflammation, such as interferon signaling, activation of EGF, nuclear factor kB, and tumor necrosis factor alpha, was able to identify HCC patients—primarily those with HCV—with poor survival after resection as well as those with cirrhosis who were more likely to develop the disease [5,99]. Another factor IL6 plays crucial roles in preventing mice from developing chemically induced HCC also plays substantial associations with the cancer hallmark [118].

Large-scale molecular and genomic investigations of HCC have shown clinically significant genomic subtypes, distinctive genomic changes associated with subtypes, and genomic predictors of these subtypes [119].

Many studies have demonstrated that targeting a member gene of a signature could improve the disease or lower the incidence rate of HCC. for example, Tyrosine kinase inhibitor gefitinib that targets the prognostic liver signature gene (Hoshida et al. 2013) member EGF's receptor, can decrease epidermal growth factor (EGF) signalling, which contributes to the development of HCC in mouse and rat models [30]. Furthermore, a gene profile highly associated with the development of HCC in HCV cirrhosis includes EGF among its top-ranked genes [5].

2.5 Drug Repositioning

The FDA (Food and Drug Administration), EMA (European Medicines Agency), and MHRA (Medicines and healthcare products regulatory agency) are all involved in the drug discovery process, which has opened the door for drug repositioning as an alternative approach. This includes the use of drugs approved by these agencies for a new indication. 31238113. The Eastern Research Group (ERG) reported that it typically takes 10-15 years to produce a new medicine, but just 2.01% of novel molecular entity developments are successful [35,120]. Drug repositioning/repositioning, commonly referred to as "old drugs for new uses," is a very effective, inexpensive, and risk-free method of identifying new indications for currently available medications. Finding novel therapeutic uses for currently available medications is referred to as drug repositioning. Investigating medications that were once approved for one application but may perhaps be beneficial for other medical ailments can be part of this. Investigating the efficacy of approved or discarded drugs for new indications using an approach called drug repositioning/purposing, can in fact overcome some of the obstacles in drug discovery, such as the necessity to meet quality standards [36-38,121]. Due to the increasing expansion of bioinformatics skills and biology big data, drug repositioning significantly minimizes the time cost of the pharmaceutical development process. In general, it takes just 1-2 years for researchers to identify new pharmacological targets and 8 years to create a repositioned drug and establishing new drug-disease interactions is the main challenge in drug repositioning [121].

The ultimate objective of repositioning research is to shorten the time and expense of medication discovery and development by properly anticipating clinical utility and using the predictions to enhance health and quality of life. As the cost and investments are lower, this provides a great opportunity for the country to develop a repositioned drug. Drug repositioning technologies seek to automate and streamline clinical trial procedures [37,40]. Additionally, another low-risk tactic is drug repositioning. The relationship between a risk and the return on investment is frequently explained using a risk-reward diagram and because repositioned drugs have been proven safe as they passed all of the phases I, II, and III clinical trials. Additionally, certain repositioned drugs might be sold as molecular entities, giving them a better chance to access the market whenever a new indication is approved [122]. Some repositioned medications cannot be sold because of this as there are several legal intellectual property (IP) and patent rights which hold a barrier to drug repositioning 30310233. Additionally, some repositioning efforts must be abandoned, which is a time and money waste (Roin, B. N. (2008). Unpatentable drugs and the standards of patentability. *Tex. L. Rev.*, *87*, 503.). For example, Sorafenib was the first targeted treatment medicine licensed by the US Food and Medicine Administration (USFDA) for advanced HCC. It slows hepatocarcinogenesis by inhibiting tumor angiogenesis and cell proliferation [123] but no medication, not even sorafenib, has been able to stop tumor recurrence following curative therapies or lower the incidence of HCC in high-risk patients as of yet which opens new avenues to discover new drugs [124]. The US FDA had authorised 72 specific anti-tumor medications by June 2016; these included the mTOR inhibitors rapamycin and everolimus, the AKT kinase inhibitors perifosine and deguelin, the RAS signalling inhibitor tipifarnib, and the PI3K inhibitor wortmannin [125].

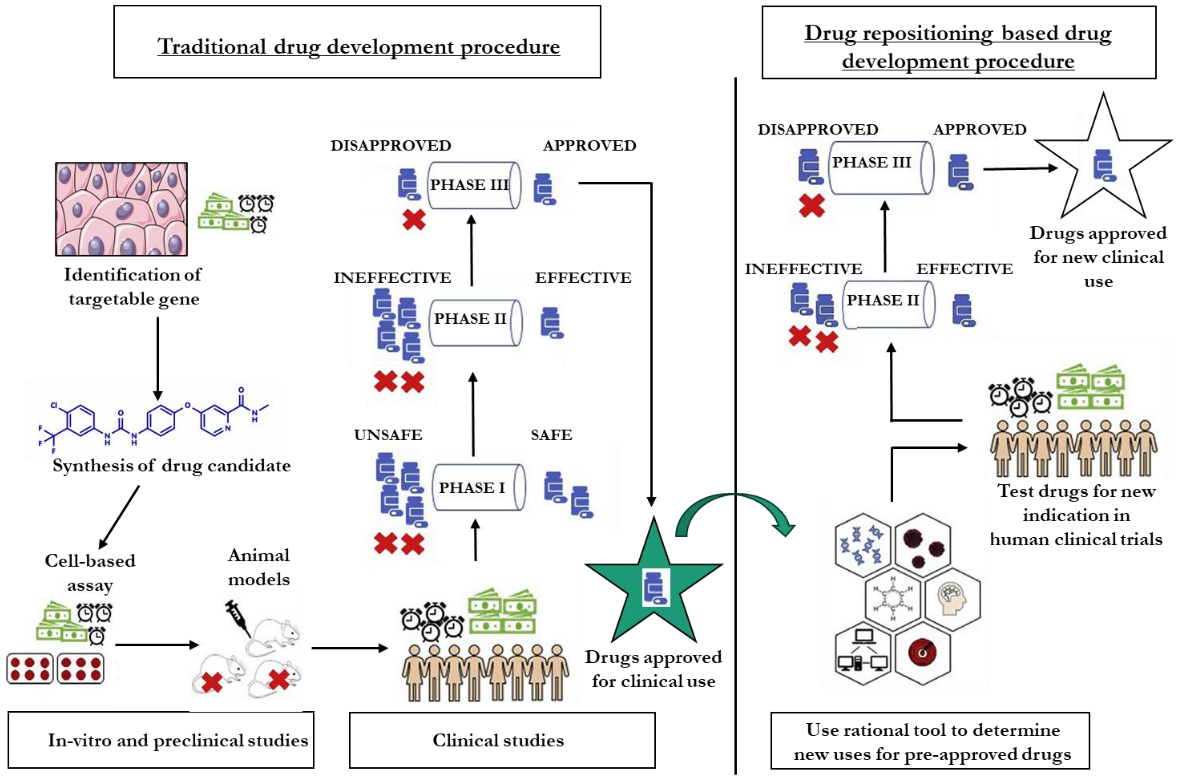


Figure 5: Overview of traditional and drug repositioning based drug development procedures

Gene signatures obtained from disease omics data with or without therapy are used in signature-based drug repositioning techniques to identify previously unrecognized off-targets or disease processes [126]. A structurally comparable medicine would have a similar target, shared biological activity, and indications. [127]. When the same metabolic pathway or signaling is damaged in both diseases, drugs that target the specific pathway can be employed in both diseases, regardless of structural dissimilarity [128,129]. In comparison to the time-consuming conventional research/development methods, drug repositioning is a less expensive and shorter strategy that provides effective therapies to patients. The presence of past knowledge of safety, efficacy, and the right administration route reduces the amount of work required while also significantly decreasing development costs and time for successfully bringing a repositioned medicine to market. This assists in overcoming the rising costs of medication research, cutting out-of-pocket expenditures for patients and, ultimately, lowering the real cost of therapy [126].

Chapter 3

Materials and Methods

3.0 Materials and Methods

3.1 HCC Signatures

A single gene expression change or a group of related changes with confirmed specificity for diagnosis, prognosis, or therapeutic response is referred to as a 'gene signature' [11]. Genes from signatures have been proved to be therapeutically targetable, in the current study, we collected genes from multiple gene signatures that involve HCC based studies. These studies show a remarkable advancement in the use of gene expression profiling to understand the molecular pathophysiology of HCC and hold the potential to enhance prognostic and diagnostic outcomes for HCC patients. Additionally, the number of genes in the signatures is sizable enough to call for a critical assessment of the validity and reproducibility of the molecular classification of HCC.

3.1.1 Prognostic signature 1

In this study, Hoshida et al. reported 186 prognostic gene signatures. For the study 360 Italian cirrhosis cohorts with various etiologies like viral hepatitis and alcohol abuse were selected. 80 non-HCV and child pugh class B cirrhosis patients were excluded from the 360-patient cohort of which 280-HCV, child pugh class A cirrhosis patients were selected. Among 280 patient samples, 44 and. 20 patient samples were excluded due to poor quality of samples and poor qualities of gene profiles respectively. Finally, the study was conducted on 216 patients having HCV and concluded that their 186 gene signature was a promising tool for the prediction of long-term clinical outcomes in patients with HCV early-stage cirrhosis. Additionally, this signature is also able to identify the patient with high risk of recurrence of HCC. In conclusion, Hoshida et al. identified 186 predictive signatures for those patients who are most in need of surveillance and strategies to prevent their development of HCC[5].

3.1.2 Prognostic signature 2

The prognosis of HCC remains poor due to high rate of recurrence and intrahepatic spread and characteristic of tumor markers depend on tumor burden, limits their values in diagnosing early-stage tumors. Therefore, Ouyang et al. identified six gene expression profiles for differentially expressed genes (DEGs) between HCC tissues and normal tissues. RNA-sequencing data of 371 HCC tissues and 50 normal tissue expressions were acquired from The Cancer Genome Atlas (TCGA). The overlapping DEGs were identified through Gene expression omnibus (GEO), Kyoto encyclopedia of genes and genome (KEGG) and Gene ontology (GO) databases. For overall survival (OS)

Univariable and Lasso-cox regression analysis was done. Multivariable Cox survival analysis was performed as well. Finally, a total of 12 gene signatures were identified to predict overall survival (OS) and risk stratification of HCC patients. For validation, 369 HCC tissues and 160 normal tissues were compared, and the analysis indicated that 12 gene prognostic signatures might be beneficial to identify low and high-risk patients and independently predict the survival of HCC patients. [24].

3.1.3 Survival signature 1

In this research article, analysis of gene expression patterns was carried out on 91 human HCC tissue and 19 normal tissues to define molecular characteristics of tumor and to find out which expression is most strongly associated with length of survival. Two distinctive subclasses of HCC were identified that exhibited a high association with patient survival. Members of 91 cases of HCC were divided in two clusters 1) Cluster A consisted of 20 chinese cohort samples showed poor OS and high AFP correlation whereas 2) Cluster B consisted of 11 chinese cohort samples and 7 Belgian cohort samples which showed better overall-survival and low AFP correlation. Analysis of those cohorts revealed a signature of 49 genes that included cell proliferation and antiapoptotic associated genes representing Cluster A, and ubiquitination and histone modification associated genes representing Cluster B. Genes representing those two clusters were found to be important discriminators of patient survival. [17].

3.1.4 Survival signature 2

This study was primarily based on two distinctive subclasses i.e Cluster A and Cluster B of HCC gene expression which were earlier reported by lee et al. (2004). In this study, 377 multidrug resistance (MDR) linked genes were analysed and it was about analysing the expression of and finding a signature of 103 DEGs in 17 normal tissues and 38 HCC tissues for comparison in two independent cohorts of patients with advanced HCC. Analysis by Gillet et al. (2016) revealed a novel 45 MDR-Linked gene signature that was able to predict overall-survival. Gillet et al. identified three small molecule inhibitors that improved OS signature of three HCC cell lines suggesting that it is possible to modulate poor-prognosis signature and chemosensitization of HCC cells can be enhanced. [22].

3.1.5 EMT signature 1

This study reported a hepatic stem cell marker, epithelial cell adhesion molecule (EpCAM) which may serve as an early biomarker of HCC onset, as its expression is highly increased in premalignant hepatic tissues. 40 HCC tissue (cohort 1) analysis showed a unique signature of EpCAM-positive expression which was identified by cDNA microarray. 238 HBV HCC tissue (cohort 2) was used for validation of the EpCAM-coexpressed gene signature. On the basis of AFP levels, EpCAM-positive and EpCAM-negative were divided into 4 groups that are (i) EpCAM+ AFP+ HCC (HpSC-HCC) (ii) EpCAM+ AFP- HCC (BDE-HCC) (iii) EpCAM- AFP+ HCC (HP-HCC) and (iv) EpCAM- AFP- (MH-HCC). This study reported 71-EMT related gene expression profiles [1].

3.1.6 EMT signature 2

The epithelial-mesenchymal transition (EMT) is crucial in cancer progression because it allows epithelial malignant cells to infiltrate and spread into nearby tissue. This study investigated 12 genes related to EMT in 128 primary HCC patients who underwent curative hepatectomy as EMT is critical for the invasiveness and metastasis of human cancer. Four-gene, CDH1, ID2 (protected genes)- downregulated and TCF3, MMP9 (risk genes)- upregulated were selected as highly predictive of survival to develop the risk score of patient survival and concluded that higher the risk score, the shorter is the overall survival (OS) and disease-free survival (DFS) [21].

3.1.7 Transcriptomic signature

The study reported transcriptome-genotype-phenotype correlation on 57 HCCs and 3 HCA (cohort 1) and validated by 63 independent HCCs tissue (cohort 2) and was achieved by QRT-PCR microarray. Boyault et al. divided tumors into 2 major groups with each further divided into 3 subgroups of HCC tumor (G1-G6) which were closely associated with genetic alterations and epigenetic deregulation. The G1 subgroup (11 gene) overexpressed genes that are expressed in fetal liver and with low viral load of HBV. The G2 subgroup (4 genes) overexpressed mutation in PIK3CA and TP53 and with high viral load of HBV. The G3 subgroup (18 genes) is characterized by mutation in TP53, and this group included high-grade tumors that are more aggressive. The G4 subgroup (1 gene) included TCF1 mutation whereas G5 (13 genes) and G6 (6 genes) were characterized by satellite nodules, higher activation of the Wnt pathway and E-cadherin down regulation[3].

3.1.8 Proliferation signature

In this study it is found that metastatic cancers from the same original site displayed unique gene expression patterns that appeared to be linked to their normal cellular progenitor. For identification of proliferation signature cDNA microarray was carried out on 102 primary HCC tissues and 74 normal tissues and 7 benign HCC. Subsequent analysis revealed that the samples expressed genes whose functions are required for cell cycle progression (specifically G2/M phase genes) and cellular proliferation rate collectively named “proliferation cluster”. Currently, the classification of the malignancies is based on primary tumor of origin, and it has significant therapy consequences. By the identification of proliferation signature, tumors of uncertain primary origin could be identified by comparing the gene expression profiles to those of other malignant and benign tissues [16].

3.1.9 Recurrence signature 1

Several vascular invasion related genes have been reported but their association with recurrence-free survival is not well known. Ming-chih Ho et al. reported that vascular invasion gene expression profile can predict the recurrence of HCC. Gene expression profile was obtained for each 18 HCC tumor samples using a microarray and reported 14 discriminative genes. The study reported a 14-gene signature which could accurately divide recurrence of HCC, indicating that the 14-gene signature can also accurately predict recurrence [18].

3.1.10 Recurrence signature 2

In this article, cirrhosis and vascular invasion, two clinicopathologic independent prognostic factors were considered at diagnosis and were combined with gene expression profiling using microarray to predict HCC recurrence. 80 HCC patients were studied into four test sets by combining the presence or absence of cirrhosis and vascular invasion. At diagnosis, patient with vascular invasion and cirrhosis were placed in subgroup 1, patient with vascular invasion and noncirrhotic condition were placed in subgroup 2, patients with no vascular invasion and cirrhotic condition placed in subgroup 3 and patient diagnosed with no vascular invasion and noncirrhotic condition were placed in subgroup 4. After 6 months they observed, subgroup 4 had no recurrence (89%-100%) which remained disease free and subgroup 1 had 78%-83% of recurrence. Subgroup 1 had a higher chance of developing recurrent disease compared to subgroup 4 when the duration of recurrence was extended. Suk Mei Wang et al. derived a 57 gene signature using a pool of 23 HCC patient samples (validation cohort) with either vascular invasion or cirrhosis (group 2/3) which could predict recurrent disease at diagnosis. An effective strategy to identify HCC recurrence with an overall accuracy above 80% was possible by the addition of 57 gene signatures with presence of vascular invasion and/or cirrhosis. If HCC's clinical course could be predicted, the best clinical therapies and aftercare plans could be put in place before the disease's recurrence manifests itself clinically [19].

3.1.11 Recurrence signature 3

Iizuka et al. investigated mRNA expression data from 33 HCC tumors as training set with use of early version of oligonucleotide microarrays with only 6000 genes. This research article predicted early intrahepatic recurrence and non-recurrence in 25 of 27 blinded samples (93%) with HCC with more accuracy than the support vector machine (SVM)-based model. The approach correctly identified early intrahepatic recurrence or nonrecurrence in 25 (93%) of 27 blinded samples, with an 88% positive predictive value and a 95% negative predictive value. This was a landmark analysis to establish the prognostic utility of genetic data from HCC tumors.

There are two types of intrahepatic recurrence: Intrahepatic metastasis and multicentric occurrence. In general, the former indicates an early recurrence after surgery and is associated with a poor prognosis. The latter is de-novo primary tumor in the liver remnant generated by ongoing virus infection and inflammation, and it typically manifests as a late recurrence. Early intrahepatic recurrence was linked to tumor size, the number of initial lesions, and vascular invasion, according to univariate analysis. Venous invasion was found to be an independent risk factor for an early intrahepatic recurrence by multivariate analysis. Norio Lizuka et al. reported a 12 gene signature pattern and among these 12 gene, TNFAIP3 and SGK gene were found to be highly downregulated in HCC with vascular invasion. They concluded that their system with the SSPR model will be of great use for identifying the metastatic potential of individual HCC and will provide new insights into bioinformatics for microarray data[20].

3.1.12 Metastatic signature

This study reported a total of 161 metastasis-related gene signatures for overall survival prediction by using 386 HCC patients after radical resection. 386 HCC cohort were divided into 2 independent, cohort 1 (247 patient) consisting of HBV+ and cohort 2 (139 patient) consisting of mixed etiologies & ethnicity analyzed with the NCI Oligo set Microarray platform. This study stated that this 161 gene signature is referred as metastasis risk classifier and is useful to predict HCC outcome and by using this metastasis risk classifier as a molecular diagnostic tool to predict the risk of having cancer relapse within 2 years of surgical resection, especially in individuals with early-stage tumors and solitary presentation. This classifier is unaffected by postoperative adjuvant therapy and can predict overall survival in patients with small solitary tumors within the first two years. Prognostic prediction of cohorts can be improved by combining AFP and metastasis risk classifiers which resulted in high risk, low risk and discordant. By evaluating primary HCC tissues in two different cohorts with mixed etiologies, Stephanie Roessler et al. validated their previously established metastasis risk classifier as a tool to predict recurrence and survival ascribed to metastatic HCC and concluded that metastasis gene classifier contributes independent prognostic value to the assessment of recurrence risk, particularly in patients with early-stage HCC where current clinical staging is unable to provide a reliable assessment [23].

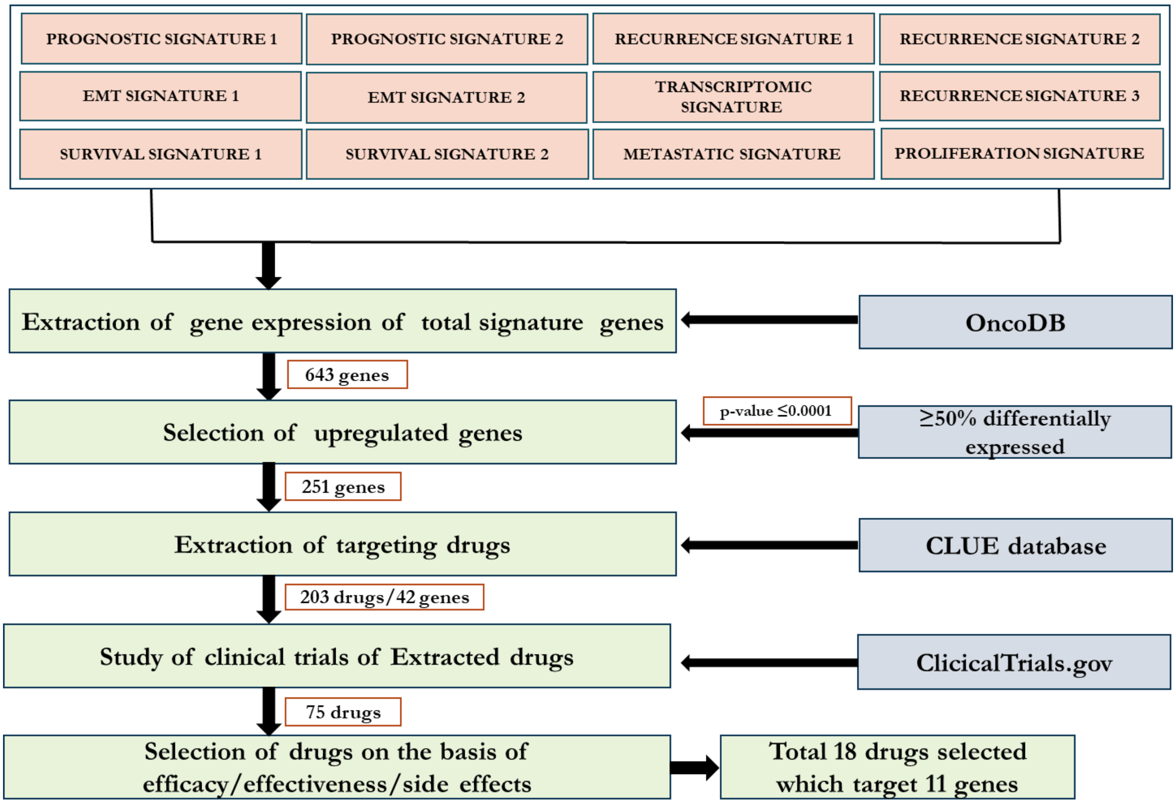


Figure 6: Gene and drug selection procedure overview

3.2 OncoDB

OncoDB is a comprehensive online database resource to investigate aberrant gene expression patterns and viral infection that are connected to cancer clinical characteristics which is a large-scale multi-omics datasets, such as those in The Cancer Genome Atlas (TCGA), like OncoDB helps us to understand the characteristics of a wide variety of tumors and comprehensive analysis of the human cancer landscape because understanding the mechanisms underlying virus-induced malignancies is made easier by the detection of oncovirus-related changes in gene expression. OncoDB helps to characterize genes based on cancer-relevant functions like fold change, gene analysis, to profile aberrant changes in tumor transcriptomes by RNA-seq analysis of different cancers. OncoDB was established to analyze functions of viral infection, a major cause of cancer samples, which has not been systematically implemented in any online tool. Differential expression analysis is frequently used to find significantly changed expression of genes between tumor and normal samples using RNA-seq data and to identify DNA methylation for epigenetic regulation to find abnormal genetic traits that underlie the development of disease can be accomplished by combining clinical data with transcriptome and epigenomic information through OncoDB. Molecular Biomarkers are generated by differentially gene expression in this way for cancer diagnosis and prognosis. These discovered RNA expression profiles have a lot of potential to be employed to predict the therapeutic response and prognosis of cancer patients. Another well-known tool is cBioportal, which concentrates on tumor samples and excludes data from normal samples for comparison analysis as well as analysis functions on DNA methylation data [130].

3.3 Extraction of expression status of gene signature members

Once we accessed OncoDB, we searched for individual genes of interest using various search criteria such as gene name, symbol, or ID. Some gene names were not recognized by the OncoDB tool, so we used Uniport and Gene card for alternative/synonyms/NCBI ID for our gene name and then successfully accessed the mRNA expression. OncoDB provides a comprehensive list of cancer genes that have been curated from various sources such as TCGA, CCLE, and COSMIC. Additionally, OncoDB provides various types of expression data such as mRNA expression, protein expression, and mutation dataand we specifically extracted mRNA expression of signature genes in normal liver tissue and HCC tumor. We employed OncoDB to run the gene analysis of desired genes in 371 HCC tumor patients and 50 normal patients.

3.4 Selection of significantly upregulated genes

After extraction of gene expression status of gene signature genes, we carried out the selection of genes based on their downregulation and upregulation. We specifically selected the genes with at least 50% alteration of mRNA expression of upregulated and downregulated genes. Here we are particularly focused on 251 upregulated genes because it is easier to inhibit an overexpressed gene using a drug rather than overexpressing a downregulated gene because it has been observed that inhibiting the gene expression suppressing factor or miRNA which degrades the mRNA of desired gene, is little difficult task as compared to simply inhibiting an upregulated gene using an inhibitor molecule. Another reason for studying only upregulated genes can be limiting the boundary of study as it requires much time and resources to work on all up/down regulated genes. After selecting upregulated genes, we validated our gene expression by evaluating its p-value which was found to be ≤ 0.0005 in all the genes.

3.5 Identification of selected genes inhibiting compounds

After selecting the upregulated genes, specific/non-specific gene targeting drugs for the selected genes were extracted using the Clue.io database which contains information on gene expression profiles and drug responses. With clue.io database can identify potential drug targets and predict how different drugs will affect different cell types. These identified drugs were specific and non-specific. We found many inhibiting compounds/drugs for one single gene and one inhibiting compound/drug for multiple genes.

3.6 Filtering of the inhibiting compounds based on current clinical trials

For finding clinical trials of each drug, we used the ClinicalTrials.gov database for studying different mechanisms of action and results of the identified drugs. We studied different clinical trials of single drugs and selected 1 clinical trial based on phase of clinical trial, used for cancerous disease, and having results. Further we selected drugs based on all-cause mortality, serious adverse event, adverse event follow-up and clinical-trial phase.

Chapter4

Result: Identification of anti-HCC therapeutically targetable genes and their drugs

4.1 Introduction

HCC gene signatures are interconnected, and they rely on several common pathways. Therefore, HCC gene signature associated studies are mainly focused on differentially expressed genes and gene targeting inhibiting molecules. HCC causative agents such as excessive alcohol consumption, HCV/HCV infection, and metabolic disorders dysregulate the lipids metabolism in hepatocytes and induced the accumulation of lipid droplets which creates metabolic pressure on hepatocytes and leads to excessive ROS and free radicals' generation and induction of apoptosis and necrosis which stimulates the inflammation and initiates fibrogenesis. A complex interplay among several genes of gene signature play which are responsible to stimulate HCC. This gene signature of Hepatocytes gradually accumulates a wide range of genetic mutations and epigenetic alterations during the development of cirrhosis, which in most cases is the cause of the onset of HCC. During this process, several risk factors that cause DNA mutations are linked to particular mutational signatures. These gene signatures are unique which are targetable and gives insight of prognosis, overall-survival and recurrence. It has been found that different subtypes of HCC cell lines showed distinct responses to molecular targeted agents. By integrating gene expression patterns and analyzing cellular origins of tumors.

4.2 Selection of upregulated genes from HCC gene signature

We studied 12 different aspects of gene expression signatures in hepatocellular carcinoma. These gene signatures were taken from different research articles given by different researchers, as shown in table 1.

These gene signatures were able to predict prognosis, recurrence, metastasis, and overall survival (Figure 6). These 12 derived gene signatures were analyzed, and a total of 643 genes were collected in this study. After choosing these 643 genes from various gene profiles, we extracted a fold change analysis of mRNA expression of the genes in HCC tissue (n=371) compared with normal tissue (n=50) samples. This extraction of mRNA expression was carried out through OncoDB, which is a large-scale multi-omics dataset, which is based on The Cancer Genome Atlas (TCGA) database. We acquired mRNA extraction and then divided genes based on their expression as differential expression. These differentially expressed genes (DEGs) include upregulated, downregulated, and less DEGs. This gene stratification was based on the criteria of genes that were equivalent to or greater than 50% differentially upregulated expressed. We specifically selected the genes with at least 50% alteration of mRNA expression of upregulated and downregulated genes. We initially collected 643 genes from a total of 12 signatures from different research articles. After extracting the gene expression, we divided 643 genes into 251 upregulated genes, 65 downregulated genes, 299 less differentially expressed genes and 29 with no data found on OncoDB. We further confirmed the significance of differential expression of gene using OncoDB, where all the genes had p-values less than 0.0005. As a result, we focused our investigation on upregulated differentially expressed genes, the majority of which were from metastatic gene signature. Here we particularly focused on upregulated gene because it is easier to inhibit an overexpressed gene with a drug than it is to overexpress a down regulated gene because it has been observed that inhibiting the gene expression suppressing factor or miRNA, which degrades the mRNA of the desired gene, is a less difficult task than simply inhibiting an upregulated gene with an inhibitor molecule. In our gene expression extraction 299 genes didn't show any expression data in OncoDB.

Table 1. Selected research articles with various gene signatures

Sr No.	Research Article Title	Signature Type	No. of Genes	DOI
1	Prognostic gene expression signature for patients with hepatitis C-related early-stage cirrhosis	Prognostic gene Signature 1	186	10.1053/j.gastro.2013.01.0
2	EpCAM and alpha-fetoprotein expression defines novel prognostic subtypes of hepatocellular carcinoma	EMT Signature 1	71	10.1158/0008-5472.CAN-07-6013
3	Transcriptome Classification of HCC is related to gene alterations and to new therapeutic targets	Transcriptomic signature	19	10.1002/hep.21467
4	Gene Expression Patterns in Human Liver Cancers	Proliferation Signature	5	10.1091/mbc.02-02-0023
5	Classification and Prediction of Survival in Hepatocellular Carcinoma by Gene Expression Profiling	Survival Signature 1	49	10.1002/hep.20375
6	A Gene Expression Profile for Vascular Invasion can Predict the Recurrence After Resection of Hepatocellular Carcinoma: a Microarray Approach	Recurrence Signature 1	14	10.1245/s10434-006-9057-
7	Identification and validation of a novel gene signature associated with the recurrence of human hepatocellular carcinoma	Recurrence Signature 2	61	10.1158/1078-0432.CCR-06-2236
8	Oligonucleotide microarray for prediction of early intrahepatic recurrence of hepatocellular carcinoma after curative resection	Recurrence Signature 3	12	10.1016/S0140-6736(03)12775-4
9	Epithelial–mesenchymal transition gene signature to predict clinical outcome of hepatocellular carcinoma	EMT Signature 2	12	10.1111/j.1349-7006.2010.01536.x
10	A Gene Expression Signature Associated with Overall Survival in Patients with Hepatocellular Carcinoma Suggests a New Treatment Strategy	Survival Signature 2	45	10.1124/mol.115.101360
11	A Unique Metastasis Gene Signature Enables Prediction of Tumor Relapse in Early Stage Hepatocellular Carcinoma Patients	Metastasis Signature	161	10.1158/0008-5472.CAN-10-2607
12	A robust twelve-gene signature for prognosis prediction of hepatocellular carcinoma	Prognostic gene Signature 2	12	10.1186/s12935-020-01294

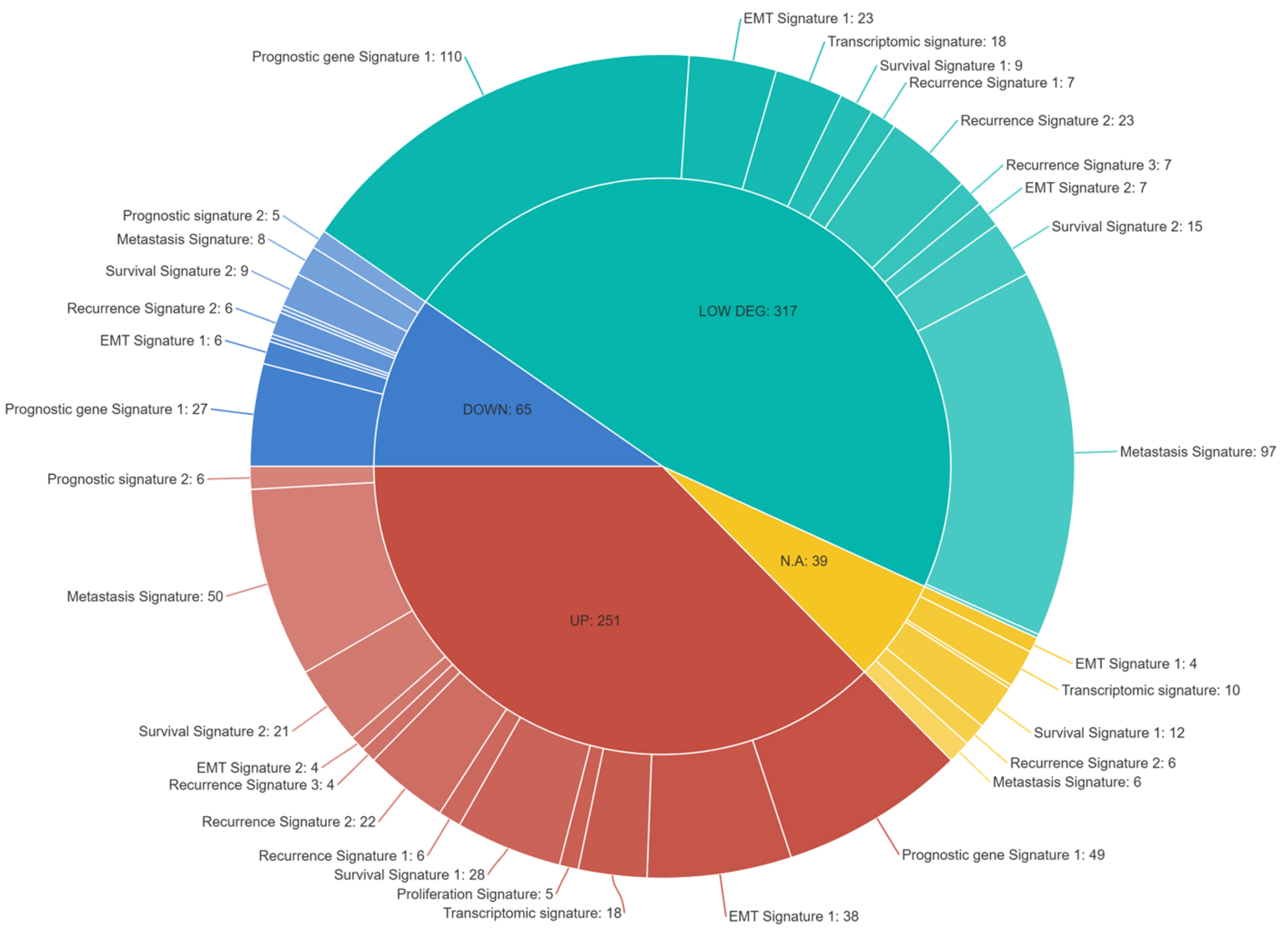


Figure 7: Gene expression status of all collected genes from several gene expression signatures

Then, using the Morpheus programme, we created a heat map of elevated genes. Morpheus is a versatile matrix visualization and analysis software that converts datasets into color boxes that are arranged in a certain grid pattern. The information is presented in a grid, with each row representing a gene and each column representing a sample. The color and intensity of the boxes are used to depict variations in gene expression (rather than absolute values). So, we created a heatmap of tumor/normal expression as well as p-values for the elevated genes.

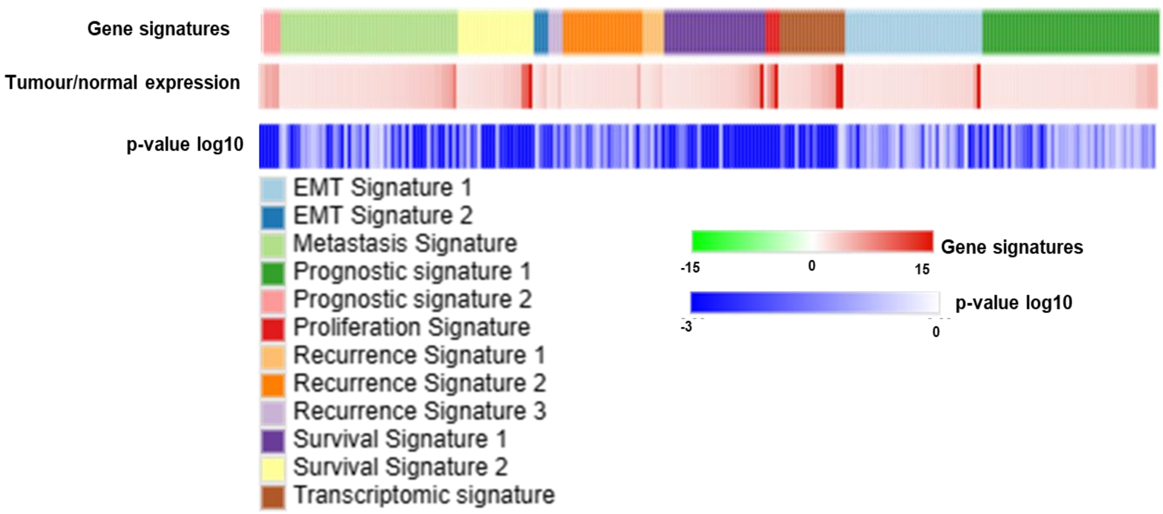


Figure 8: Heatmap representation of expression of differentially upregulated genes with ≥50% gene expression in HCC tumor samples as compared to normal liver samples.

4.3 Extraction of molecular inhibitors for selected upregulated genes

Using the Clue.io database, which provides information on gene expression profiles and pharmacological responses, specific/non-specific gene targeting drugs for the identified genes. With the clue.io database, it is possible to find prospective pharmacological targets and predict how different drugs could affect different cell types. As a result, the selected elevated genes were examined for the presence of gene-targeted drug/inhibitor compounds. Only 42 of the 251 elevated genes had specific/non-specific gene targeting drug/inhibitor compounds. We identified a total 203 gene targeting drugs /inhibitors for 42 upregulating genes and remaining 209 genes did not show any targeting inhibitor (Figure 8). These 42 genes each have multiple targeted inhibitor compounds, for a total of 203 inhibitor molecules discovered. These 203 inhibitor molecules were specific/non-specific gene targeting molecules and one inhibitor targeted one gene whereas another targeted more than one gene, so we utilized a sankey plot to determine the degree of overlap between drugs/inhibitor molecule and genes. This sankey plot involved overlapping of 203 drugs of 42 genes of 12 gene signatures. It was observed that 12 gene signatures also shared common genes among them. For example, MMP9 gene appeared in metastatic signature, EMT signature, and survival signature. Another example, TTK gene appeared in prognostic signature, EMT signature, survival signature, and transcriptomic signature.

Similarly, many genes share common drugs, for example dasatinib drug targets ABL2 of metastatic signature and DDR1 of prognostic gene signature, puromycin and anisomycin drugs target both RPL23A and RPL13A genes of EMT signature. Likewise, BX-912 and BX-795 drugs target both PDK1 and CHEK1 genes of metastatic signature. ENMD-2076 drug targets AURKA and SRC survival signature. SB-205580 drug targets CHEK1 and SRC genes of survival signature. Alvocidib (Flavopiridol) drug targeted CDK4 and BIRC5 gene of survival signature and SB-216763 drug targets CCNA2 gene which is present in transcriptomic signature and survival signature 1.

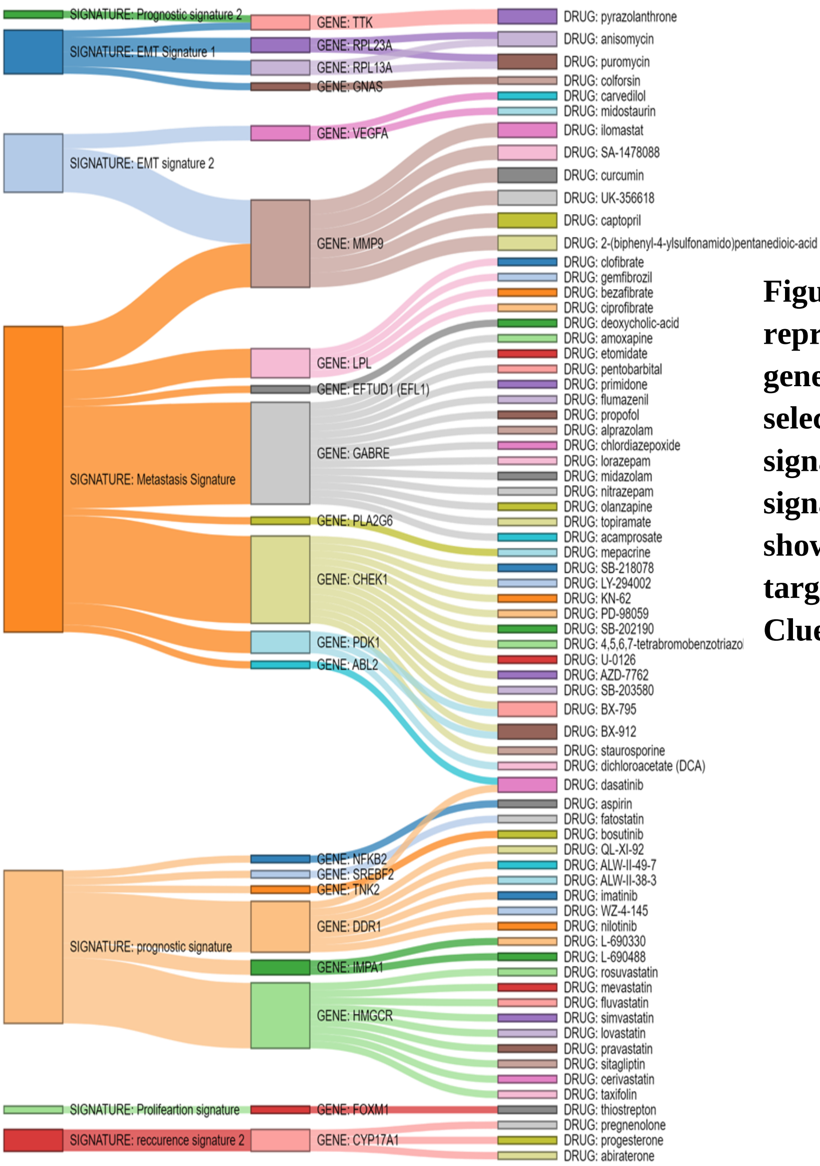
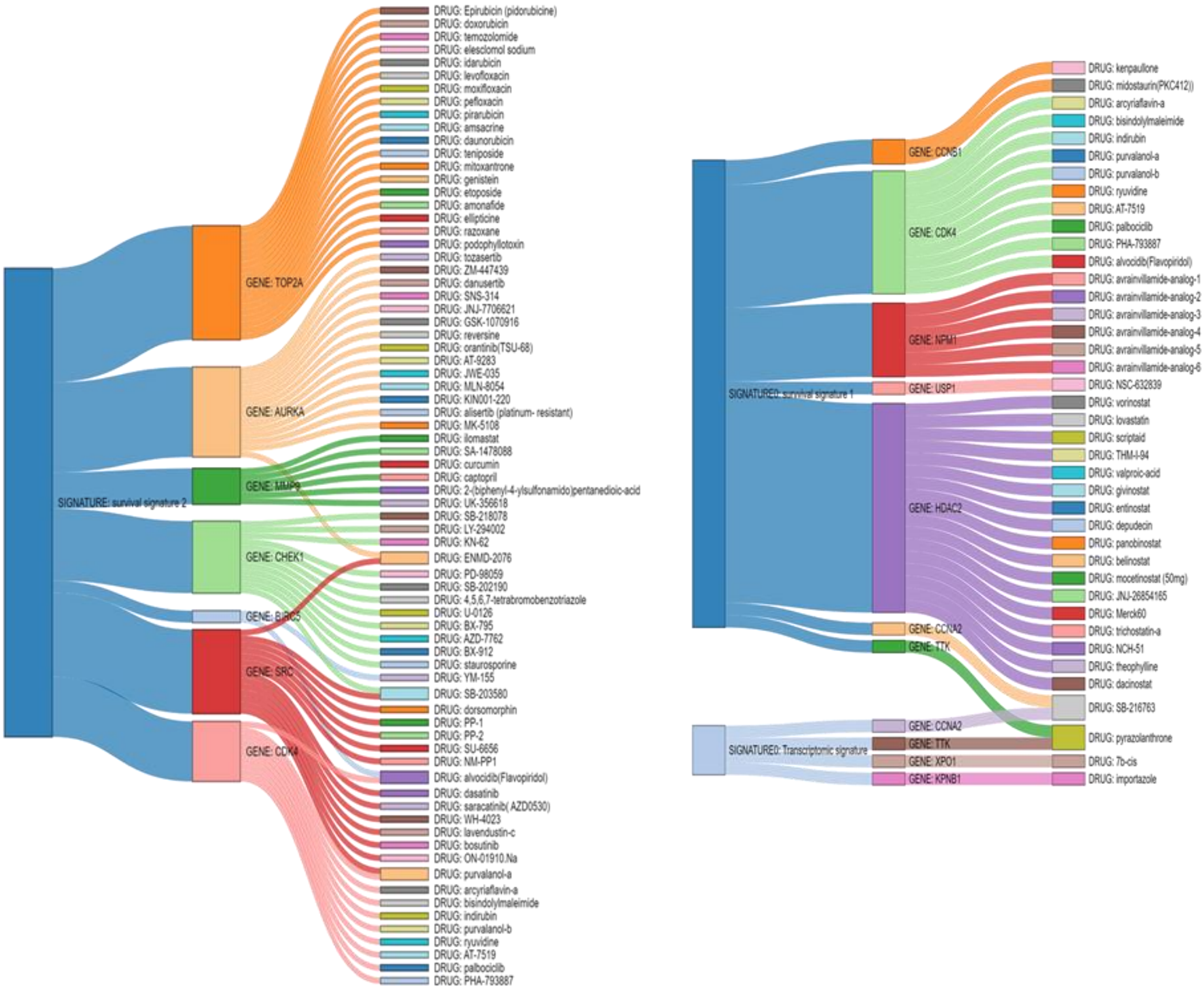


Figure 9: Sanky plots representing the expression of gene signature members of all selected genes except recurrence signature 1 and recurrence signature 3. Boxes on the right showing the selected gene targeting drugs extracted from Clue.io database.



4.4 Retrieving and selection of clinical trials involving extracted inhibitor molecules

We used the ClinicalTrials.gov database to discover clinical trials for each drug to explore the various mechanisms of action and outcomes of the selected treatments. We looked at numerous clinical trials for a single medicine and choose one based on the phase of the trial, whether it was for a malignant illness, and the outcome of the experiment. As a result, we reviewed all 203 drugs that were previously taken from the Clue.io database again. 75 out of the 203 drugs were had completed clinical trial studies with results (Figure 9,A). Other criteria of selecting a particular clinical trial were that malignant illness had no additional drug interventions so clinical trials with the specific drug effect were selected.

The clinicalTrial.gov database was searched for clinical studies using drug names, and many clinical trials were found which are already completed. Initially, the selection of clinical trials must have results after that priority in clinical trial selection was given to a single drug's induced action mechanism, and clinical trials having additional drugs with the extracted drug was not considered (except placebo). For example, a drug named simvastatin has a total of 164 clinical studies with results out of which, studies with additional drug results were not considered. Following that, we looked for clinical trials of malignant illnesses that were similar to HCC. Several malignant illnesses have been diagnosed, and clinical phase investigations are being considered. Drugs with no extra drug outcomes and for diverse malignant conditions in the same trial phase were then chosen based on all-cause mortality, serious adverse events, and adverse event follow up. Less all-cause mortality, lower serious adverse event rates, and high adverse event follow-up were all favourable for drug selection in numerous clinical studies. For examples here the clinical trial selected for simvastatin was NCT00334542. Medical condition was breast cancer and all-cause mortality is 0, serious adverse event is 0, adverse event follow up is 0 and clinical trial phase is 2. Similarly, all the 75 drugs were subjected for the study of clinical trials. After collecting the all-cause mortality, serious adverse event, adverse event, and clinical phase trial data of all 75 drugs, a heat map was generated using Morpheus software.

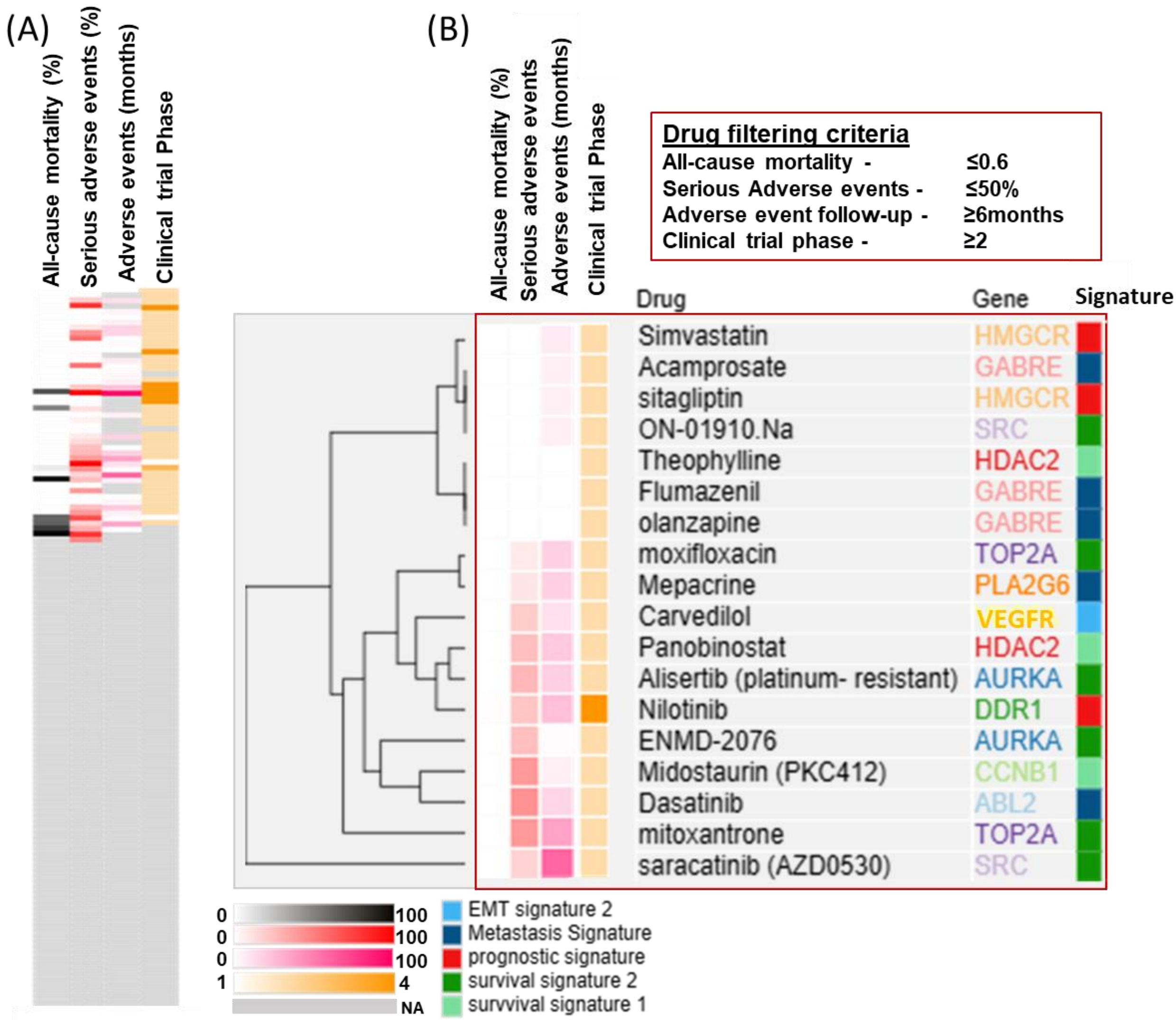
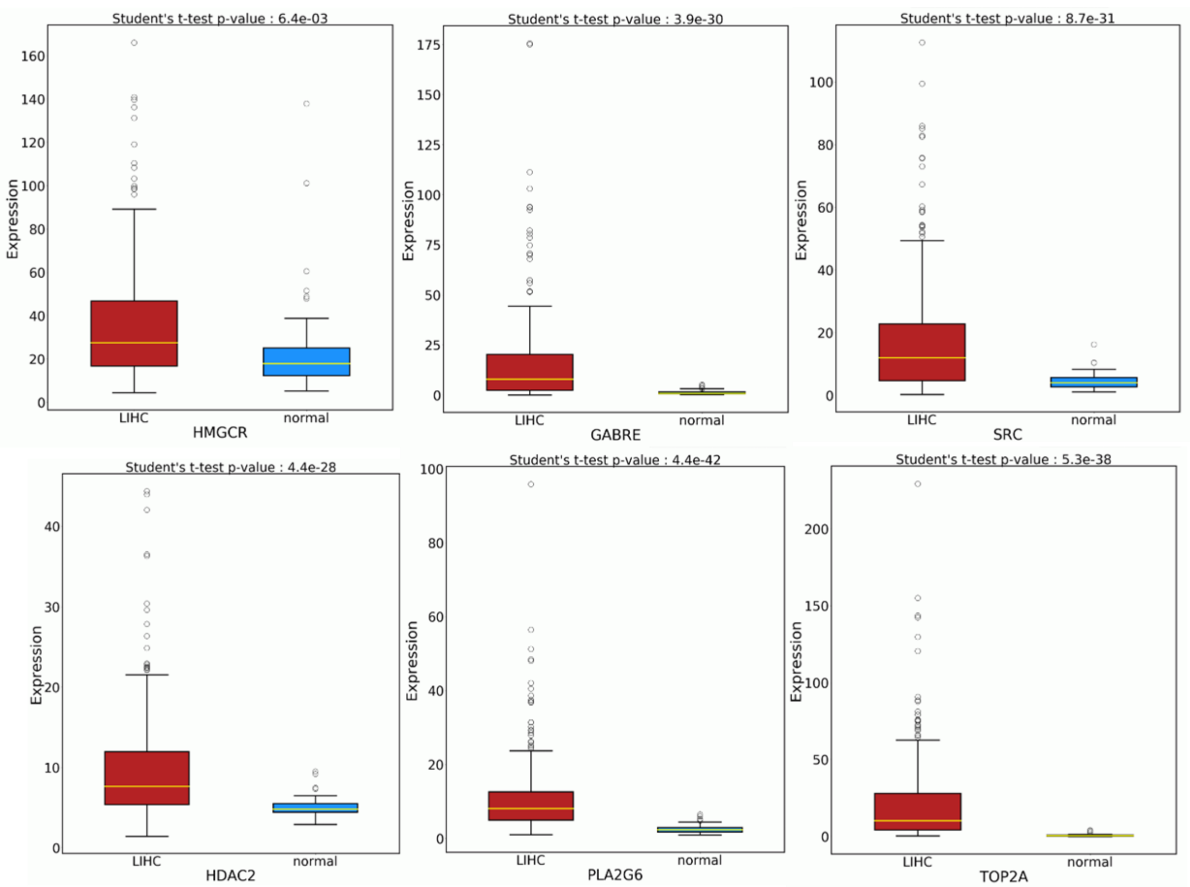


Figure 10: Selection of drugs. (A) Drugs with available clinical trials (B) Selected drugs on the basis of clinical trials' outcomes.

4.5 Filtering of inhibitor molecules and their genes based on clinical trials outcomes.

A heatmap was generated of 75 drugs with clinical trials. These 75 drugs must be sorted by all-cause mortality, significant adverse event, adverse event, and phase. We retained a drug filtering strategy in place to narrow down our findings for anti-HCC potential drugs. Candidate drugs were chosen based on drug filtering criteria, and the other drugs were discarded. The drug filtering criteria was as, all-cause mortality should be ≤0.6, serious adverse event should be ≤50%, adverse event follow up should be ≥6 months and clinical phase trial should be ≥2. 18 candidate drugs were selected based on drug filtering criteria of 75 drugs with clinical trials(Figure 9,B). Additionally, this 18-candidate drug heat map was generated and hierarchical clustering with Euclidean distance was applied. For most common hierarchical clustering software, the default distance measure is the Euclidean distance. Genes with similar expression patterns are grouped together and are connected by a series of branches (clustering tree or dendrogram) when hierarchical clustering. A total 18 drugs targeting 11 gene are Simvastatin (NCT00334542) and sitagliptin (NCT01963845) drugs targeting HMGCR gene, Acamprosate (NCT04287920), Flumazenil (NCT01183312) and olanzapine (NCT03137121) drugs targeting GABRE gene, ON-01910.Na (NCT00856791) and Saracatinib (NCT00607594) drugs targeting SRC gene, Theophylline (NCT02463409) and Panobinostat (NCT01802879) drugs targeting HDAC2 gene, Moxifloxacin (NCT00082173) and Mitoxantrone (NCT00477087) drugs targeting TOP2A, Mepacrine (NCT00417274) drug targeting PLA2G6, Carvedilol (NCT02177175) drug targeting VEGFR gene, Alisertib (platinum-resistant) (NCT00853307) and ENMD-2076 (NCT01914510) drugs targeting AURKA, Nilotinib (NCT02546674) drug targeting DDR1 gene, Midostaurin (NCT00866281) drug targeting CCNB1 gene and Dasatinib (NCT00858403) drug targeting ABL2 gene.



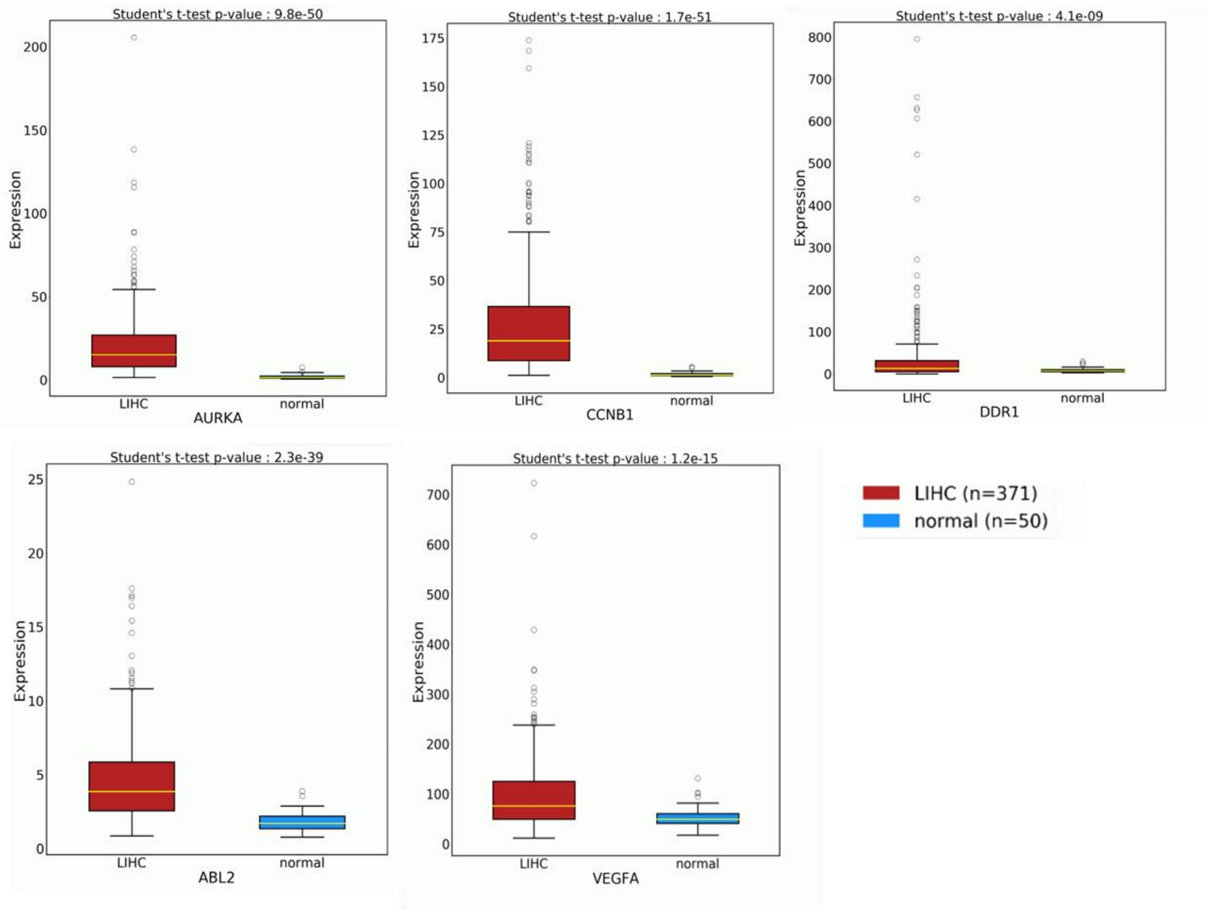


Figure 11: Expression of selected genes in HCC tumor samples and normal liver samples.

4.6 Assessment of selected genes for their role in disease free survival

Alteration in expression of one or more than one gene from the above-mentioned identified 11 genes may exert a role in disease-free survival of patients. To assess this hypothesis, we evaluated the role of the identified genes on disease-free survival after tumor resection with the help of cBioportal database. Disease- free survival curve indicated that the patient having altered group (red curve) showed decreased in disease-free survival after tumor resection whereas unaltered group (blue curve) showed less decreased in disease-free survival compared with altered group indicating great disease-free survival in unaltered group. Further, we were interested in checking the alteration in gene expression of the identified genes in different ethnicities and we found that the Asian and white population showed maximum alteration while the black or African American group showed very low alteration of those identified genes (Figure 11).

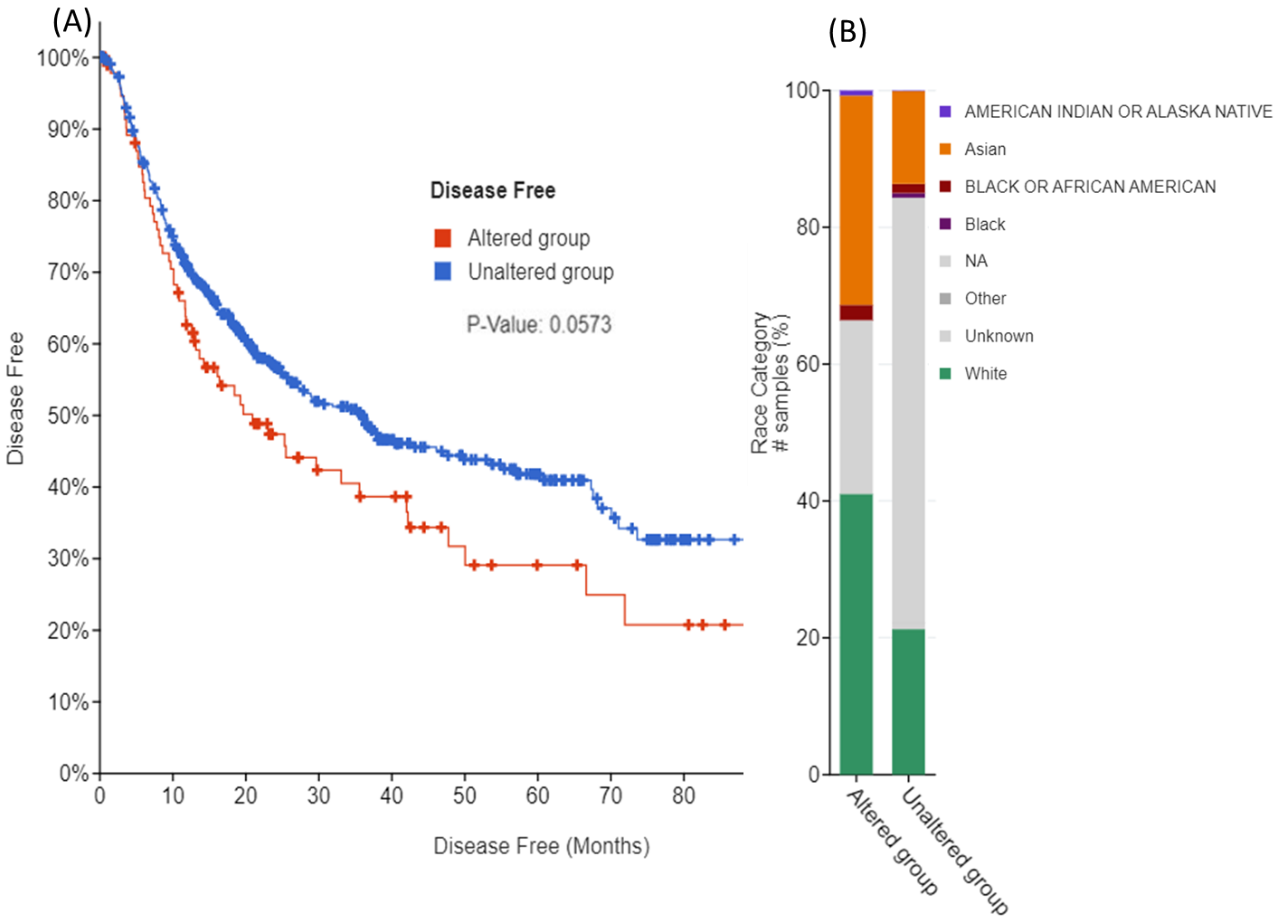


Figure 12: Effect of selected genes' alteration in patients (A) Effect of gene alteration on the diseases free survival of patients after tumor resection, (B) Selected gene alteration in patients from different ethnic groups.

Above results validate our hypothesis that selected targetable genes are involved in disease-free survival of the patient where we can use these 18 candidate drugs targeting 11 genes for drug repositioning in hepatocellular carcinoma.

Chapter 5

Discussion and Conclusion

In this study we analyzed the gene expression of the collected genes (643 genes) from the gene signature of different aspects of HCC. This gene mutation or alteration was responsible for the development of HCC and this gene expression signature could provide insight about the overall survival of the patient, metastasis, proliferation, prognosis of HCC and recurrence of HCC. These gene signatures hold a high potential value for targeting molecules to inhibit the cascade pathway of different aspects of HCC.

A set of differentially expressed genes was generated from which particularly focused on upregulated differentially expressed genes (251 genes). Using these 251 genes and employing drug repositioning could be a better option to find interventions and validate using initial in-vivo/in-vitro experiments which makes it easier to analyze, saves time and is a cost-effective strategy for finding therapeutic drugs for HCC.

To use this strategy, we extracted molecules that could inhibit or alter the upregulated gene expression and we found a total of 203 drugs which were inhibiting 42 genes out of 251 upregulated genes. Most of the drugs, 128 are not clinically tested yet. We were specifically interested in remaining drugs having clinical trials, a total of 75 drugs. The clinical trials selected for these75 drugs were different malignant illnesses (except HCC) and so these drugs can be used for repositioning as they are the same gene targeting drug. For the selection of a drug to be repositioned the following parameters like all-cause mortality should be nearly 0, serious-adverse events should be moderately low and adverse event follow-up should be quite long and higher clinical phase should be followed by the drug.

These 75 drugs held a potential for being repositioned but by seeing their efficacy, effectiveness, and side effects most of them were eliminated through the drug selection process. We screened these 75 medications using the following criteria: all-cause mortality should equal to or less than 0.6, serious adverse events should be equal to or less than 50%, adverse event follow-up should equal to or more than 6 months, and clinical phase trial should equal to or less than 2. 18 drugs were able to pass these criteria making them potential to be repositioned because of their efficacy, effectiveness, and side effects. These 18 drugs targeted a total of 11 genes that are being altered in HCC. So here we Identified 18 anti-Hepatocellular Carcinoma candidate drugs for repositioning using gene signatures.

The disease-free survival curve validated the alteration of the 11 gene in HCC and disease-free survival increases if the alteration was inhibited by these 18 anti-HCC drugs. The altered group of gene showed decrease in survival-free curve while unaltered group of gene showed less decrease in comparison to altered genes. Further this alteration was high in Asian and white group ethnicities while the black or African American group showed very less alteration in these 11 genes. This provides an insight for the therapeutic indication that which ethnicity is more likely to be benefited with the selected candidate drugs. The Asian and white group are more likely to benefit and show a better prognosis.

These 18 candidate drugs can be further tested for in-vitro validations on HCC cell lines through multiple cell assays. This candidate drug should inhibit one of the cancers associated properties like proliferation, migration, angiogenesis, invasion, and colony formation. If any of the drugs inhibit one of the properties, then it has a strong therapeutic potential against HCC. The drug effect could be tested through MTT/Glo2.0 assay for proliferation where the drug must inhibit the proliferation of the cell indirectly inhibiting cell cycle and the genes for cell cycle regulation are HDAC2, SRC, CCNB1, VEGFR, TOP2A, ABL2 and AURKA. The scratch migration assay for migration where the drug must act as anti-migratory drug and the gene for migration is ABL2, DDR1 and VEGFR. The Matrigel invasion chamber assay for invasion. The tube formation assay for angiogenic properties should be inhibited and the gene responsible is VEGFR. Potent drugs can further be evaluated using in-vivo models such as diet/alcohol/toxin/drug induced models of HCC.

Table 2: List of selected genes, their functions/targeting drugs and mechanism of action.

Sr. No.	GENE	FUNCTION OF GENE	DRUG	MECHANISM OF ACTION
1	HMGCR	Plays a critical role in cellular cholesterol homeostasis, HMGCR is the main target of statins, a class of cholesterol-lowering drugs	Simvastatin, sitagliptin	HMGCR inhibitor
2	GABRE	GABA, the major inhibitory neurotransmitter in the vertebrate brain, mediates neuronal inhibition by binding to the GABA/benzodiazepine receptor and opening an integral chloride channel.	Flumazenil, Acamprosate, olanzapine	GABRE inhibitor
3	SRC	Plays an important role in the regulation of cytoskeletal organization through phosphorylation of specific substrates. Has a critical role in the stimulation of the CDK20/MAPK3 mitogen-activated protein kinase cascade by epidermal growth factor (Probable)	ON-01910.Na, saracatinib (AZD0530)	SRC inhibitor
4	HDAC2	Histone deacetylase that catalyzes the deacetylation of lysine residues on the N-terminal part of the core histones (H2A, H2B, H3 and H4)	Theophylline, Panobinostat	HDAC2 inhibitor
5	TOP2A	Key decatenating enzyme that alters DNA topology by binding to two double-stranded DNA molecules, generating a double-stranded break in one of the strands, passing the intact strand through the broken strand, and religating the broken strand	moxifloxacin, mitoxantrone	TOP2A inhibitor
6	PLA2G6	Calcium-independent phospholipase involved in phospholipid remodeling with implications in cellular membrane homeostasis, mitochondrial integrity and signal transduction.	Mepacrine	PLA2G6 inhibitor

7	VEGFA	Growth factor active in angiogenesis, vasculogenesis and endothelial cell growth. Induces endothelial cell proliferation, promotes cell migration, inhibits apoptosis and induces permeabilization of blood vessels.	Carvedilol	VEGFA inhibitor
8	AURKA	Mitotic serine/threonine kinase that contributes to the regulation of cell cycle progression	ENMD-2076, Alisertib (platinum-resistant)	AURKA inhibitor
9	CCNB1	Essential for the control of the cell cycle at the G2/M (mitosis) transition	Midostaurin (PKC412)	CCNB1 inhibitor
10	ABL2	Non-receptor tyrosine-protein kinase that plays an ABL1-overlapping role in key processes linked to cell growth and survival such as cytoskeleton remodeling in response to extracellular stimuli, cell motility and adhesion and receptor endocytosis.	Dasatinib	ABL2 inhibito
11	DDR1	Tyrosine kinase that functions as cell surface receptor for fibrillar collagen and regulates cell attachment to the extracellular matrix, remodeling of the extracellular matrix, cell migration, differentiation, survival and cell proliferation. Collagen binding triggers a signaling pathway that involves SRC and leads to the activation of MAP kinases	Nilotinib	DDR1 inhibitor

Chapter 6

References

References:

1. Yamashita T, Forgues M, Wang W, et al. EpCAM and alpha-fetoprotein expression defines novel prognostic subtypes of hepatocellular carcinoma. *Cancer Res.* 2008;68(5):1451-1461.

2. Chidambaranathan-Reghupaty S, Fisher PB, Sarkar D. Hepatocellular carcinoma (HCC): Epidemiology, etiology and molecular classification. *Adv Cancer Res.* 2021;149:1-61.

3. Boyault S, Rickman DS, de Reynies A, et al. Transcriptome classification of HCC is related to gene alterations and to new therapeutic targets. *Hepatology.* 2007;45(1):42-52.

4. Llovet JM, Zucman-Rossi J, Pikarsky E, et al. Hepatocellular carcinoma. *Nat Rev Dis Primers.* 2016;2:16018.

5. Hoshida Y, Villanueva A, Sangiovanni A, et al. Prognostic gene expression signature for patients with hepatitis C-related early-stage cirrhosis. *Gastroenterology.* 2013;144(5):1024-1030.

6. Wu Y, Liu Z, Xu X. Molecular subtyping of hepatocellular carcinoma: A step toward precision medicine. *Cancer Commun (Lond).* 2020;40(12):681-693.

7. Papatheodoridis GV, Idilman R, Dalekos GN, et al. The risk of hepatocellular carcinoma decreases after the first 5 years of entecavir or tenofovir in Caucasians with chronic hepatitis B. *Hepatology.* 2017;66(5):1444-1453.

8. Fujiwara N, Friedman SL, Goossens N, Hoshida Y. Risk factors and prevention of hepatocellular carcinoma in the era of precision medicine. *J Hepatol.* 2018;68(3):526-549.

9. McGlynn KA, Petrick JL, El-Serag HB. Epidemiology of Hepatocellular Carcinoma. *Hepatology.* 2021;73 Suppl 1(Suppl 1):4-13.

10. Li Q, Cao M, Lei L, et al. Burden of liver cancer: From epidemiology to prevention. *Chin J Cancer Res.* 2022;34(6):554-566.

11. Chibon F. Cancer gene expression signatures - the rise and fall? *Eur J Cancer.* 2013;49(8):2000-2009.

12. Oldenhuis CN, Oosting SF, Gietema JA, de Vries EG. Prognostic versus predictive value of biomarkers in oncology. *Eur J Cancer.* 2008;44(7):946-953.

13. Nguyen HG, Welty CJ, Cooperberg MR. Diagnostic associations of gene expression signatures in prostate cancer tissue. *Curr Opin Urol.* 2015;25(1):65-70.

14. Baker SG, Kramer BS. Evaluating surrogate endpoints, prognostic markers, and predictive markers: Some simple themes. *Clin Trials.* 2015;12(4):299-308.

15. Warner DF. Defining a diagnostic gene signature for tuberculosis. *Lancet Respir Med.* 2016;4(3):170-171.

16. Chen X, Cheung ST, So S, et al. Gene expression patterns in human liver cancers. *Mol Biol Cell.* 2002;13(6):1929-1939.

17. Lee JS, Chu IS, Heo J, et al. Classification and prediction of survival in hepatocellular carcinoma by gene expression profiling. *Hepatology.* 2004;40(3):667-676.

18. Ho MC, Lin JJ, Chen CN, et al. A gene expression profile for vascular invasion can predict the recurrence after resection of hepatocellular carcinoma: a microarray approach. *Ann Surg Oncol.* 2006;13(11):1474-1484.

19. Wang SM, Ooi LL, Hui KM. Identification and validation of a novel gene signature associated with the recurrence of human hepatocellular carcinoma. *Clin Cancer Res.* 2007;13(21):6275-6283.

20. Iizuka N, Oka M, Yamada-Okabe H, et al. Oligonucleotide microarray for prediction of early intrahepatic recurrence of hepatocellular carcinoma after curative resection. *Lancet.* 2003;361(9361):923-929.

21. Kim J, Hong SJ, Park JY, et al. Epithelial-mesenchymal transition gene signature to predict clinical outcome of hepatocellular carcinoma. *Cancer Sci.* 2010;101(6):1521-1528.

22. Gillet JP, Andersen JB, Madigan JP, et al. A Gene Expression Signature Associated with Overall Survival in Patients with Hepatocellular Carcinoma Suggests a New Treatment Strategy. *Mol Pharmacol.* 2016;89(2):263-272.

23. Roessler S, Jia HL, Budhu A, et al. A unique metastasis gene signature enables prediction of tumor relapse in early-stage hepatocellular carcinoma patients. *Cancer Res.* 2010;70(24):10202-10212.

24. Ouyang G, Yi B, Pan G, Chen X. A robust twelve-gene signature for prognosis prediction of hepatocellular carcinoma. *Cancer Cell Int.* 2020;20:207.

25. Nault JC, Martin Y, Caruso S, et al. Clinical Impact of Genomic Diversity From Early to Advanced Hepatocellular Carcinoma. *Hepatology.* 2020;71(1):164-182.

26. Lee SC, Tan HT, Chung MC. Prognostic biomarkers for prediction of recurrence of hepatocellular carcinoma: current status and future prospects. *World J Gastroenterol.* 2014;20(12):3112-3124.

27. Trost B, Moir CA, Gillespie ZE, Kusalik A, Mitchell JA, Eskiw CH. Concordance between RNA-sequencing data and DNA microarray data in transcriptome analysis of proliferative and quiescent fibroblasts. *R Soc Open Sci.* 2015;2(9):150402.

28. Lu J, Chen Y, Zhang X, Guo J, Xu K, Li L. A novel prognostic model based on single-cell RNA sequencing data for hepatocellular carcinoma. *Cancer Cell Int.* 2022;22(1):38.

29. Iizuka N, Hamamoto Y, Tsunedomi R, Oka M. Translational microarray systems for outcome prediction of hepatocellular carcinoma. *Cancer Sci.* 2008;99(4):659-665.

30. Schiffer E, Housset C, Cacheux W, et al. Gefitinib, an EGFR inhibitor, prevents hepatocellular carcinoma development in the rat liver with cirrhosis. *Hepatology.* 2005;41(2):307-314.

31. Qian T, Fujiwara N, Koneru B, et al. Molecular Signature Predictive of Long-Term Liver Fibrosis Progression to Inform Antifibrotic Drug Development. *Gastroenterology.* 2022;162(4):1210-1225.

32. Ye QH, Qin LX, Forgues M, et al. Predicting hepatitis B virus-positive metastatic hepatocellular carcinomas using gene expression profiling and supervised machine learning. *Nat Med.* 2003;9(4):416-423.

33. Sun BS, Dong QZ, Ye QH, et al. Lentiviral-mediated miRNA against osteopontin suppresses tumor growth and metastasis of human hepatocellular carcinoma. *Hepatology.* 2008;48(6):1834-1842.

34. Takafuji V, Forgues M, Unsworth E, Goldsmith P, Wang XW. An osteopontin fragment is essential for tumor cell invasion in hepatocellular carcinoma. *Oncogene.* 2007;26(44):6361-6371.

35. Brodniewicz T, Grynkiewicz G. Preclinical drug development. *Acta Pol Pharm.* 2010;67(6):578-585.

36. Parisi D, Adasme MF, Sveshnikova A, Bolz SN, Moreau Y, Schroeder M. Drug repositioning or target repositioning: A structural perspective of drug-target-indication relationship for available repurposed drugs. *Comput Struct Biotechnol J.* 2020;18:1043-1055.

37. Sleire L, Forde HE, Netland IA, Leiss L, Skeie BS, Enger PO. Drug repurposing in cancer. *Pharmacol Res.* 2017;124:74-91.

38. Huang B, Zhang Y. Teaching an old dog new tricks: Drug discovery by repositioning natural products and their derivatives. *Drug Discov Today.* 2022;27(7):1936-1944.

39. Aggarwal S, Verma SS, Aggarwal S, Gupta SC. Drug repurposing for breast cancer therapy: Old weapon for new battle. *Semin Cancer Biol.* 2021;68:8-20.

40. Schuler J, Falls Z, Mangione W, Hudson ML, Bruggemann L, Samudrala R. Evaluating the performance of drug-repurposing technologies. *Drug Discov Today.* 2022;27(1):49-64.

41. Pushpakom S, Iorio F, Eyers PA, et al. Drug repurposing: progress, challenges and recommendations. *Nat Rev Drug Discov.* 2019;18(1):41-58.

42. Jourdan JP, Bureau R, Rochais C, Dallemagne P. Drug repositioning: a brief overview. *J Pharm Pharmacol.* 2020;72(9):1145-1151.

43. Iwata H, Sawada R, Mizutani S, Yamanishi Y. Systematic drug repositioning for a wide range of diseases with integrative analyses of phenotypic and molecular data. *J Chem Inf Model.* 2015;55(2):446-459.

44. Sung H, Ferlay J, Siegel RL, et al. Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries. *CA Cancer J Clin.* 2021;71(3):209-249.

45. Llovet JM, Kelley RK, Villanueva A, et al. Hepatocellular carcinoma. *Nat Rev Dis Primers.* 2021;7(1):6.

46. Marrero JA, Kulik LM, Sirlin CB, et al. Diagnosis, Staging, and Management of Hepatocellular Carcinoma: 2018 Practice Guidance by the American Association for the Study of Liver Diseases. *Hepatology.* 2018;68(2):723-750.

47. Schulze K, Imbeaud S, Letouze E, et al. Exome sequencing of hepatocellular carcinomas identifies new mutational signatures and potential therapeutic targets. *Nat Genet.* 2015;47(5):505-511.

48. Letouze E, Shinde J, Renault V, et al. Mutational signatures reveal the dynamic interplay of risk factors and cellular processes during liver tumorigenesis. *Nat Commun.* 2017;8(1):1315.

49. Chakraborty E, Sarkar D. Emerging Therapies for Hepatocellular Carcinoma (HCC). *Cancers (Basel).* 2022;14(11).

50. Fattovich G, Stroffolini T, Zagni I, Donato F. Hepatocellular carcinoma in cirrhosis: incidence and risk factors. *Gastroenterology.* 2004;127(5 Suppl 1):S35-50.

51. El-Serag HB, Rudolph KL. Hepatocellular carcinoma: epidemiology and molecular carcinogenesis. *Gastroenterology.* 2007;132(7):2557-2576.

52. Frezza M, di Padova C, Pozzato G, Terpin M, Baraona E, Lieber CS. High blood alcohol levels in women. The role of decreased gastric alcohol dehydrogenase activity and first-pass metabolism. *N Engl J Med.* 1990;322(2):95-99.

53. Naveau S, Giraud V, Borotto E, Aubert A, Capron F, Chaput JC. Excess weight risk factor for alcoholic liver disease. *Hepatology.* 1997;25(1):108-111.

54. Hart CL, Morrison DS, Batty GD, Mitchell RJ, Davey Smith G. Effect of body mass index and alcohol consumption on liver disease: analysis of data from two prospective cohort studies. *BMJ.* 2010;340:c1240.

55. Raynard B, Balian A, Fallik D, et al. Risk factors of fibrosis in alcohol-induced liver disease. *Hepatology.* 2002;35(3):635-638.

56. Kane SF, Roberts C, Paulus R. Hereditary Hemochromatosis: Rapid Evidence Review. *Am Fam Physician.* 2021;104(3):263-270.

57. Czlonkowska A, Litwin T, Dusek P, et al. Wilson disease. *Nat Rev Dis Primers.* 2018;4(1):21.

58. Strnad P, McElvaney NG, Lomas DA. Alpha(1)-Antitrypsin Deficiency. *N Engl J Med.* 2020;382(15):1443-1455.

59. Dragani TA. Risk of HCC: genetic heterogeneity and complex genetics. *J Hepatol.* 2010;52(2):252-257.

60. El-Serag HB. Epidemiology of viral hepatitis and hepatocellular carcinoma. *Gastroenterology.* 2012;142(6):1264-1273 e1261.

61. Romano A, Angeli P, Piovesan S, et al. Newly diagnosed hepatocellular carcinoma in patients with advanced hepatitis C treated with DAAs: A prospective population study. *J Hepatol.* 2018;69(2):345-352.

62. Chen T, Liu J, Li Y, Wei S. Burden of Disease Associated with Dietary Exposure to Aflatoxins in China in 2020. *Nutrients.* 2022;14(5).

63. Fu J, Wang H. Precision diagnosis and treatment of liver cancer in China. *Cancer Lett.* 2018;412:283-288.

64. Jinjuvadia R, Patel S, Liangpunsakul S. The association between metabolic syndrome and hepatocellular carcinoma: systemic review and meta-analysis. *J Clin Gastroenterol.* 2014;48(2):172-177.

65. Benvegnu L, Gios M, Boccato S, Alberti A. Natural history of compensated viral cirrhosis: a prospective study on the incidence and hierarchy of major complications. *Gut.* 2004;53(5):744-749.

66. Fattovich G, Pantalena M, Zagni I, et al. Effect of hepatitis B and C virus infections on the natural history of compensated cirrhosis: a cohort study of 297 patients. *Am J Gastroenterol.* 2002;97(11):2886-2895.

67. Bosch FX, Ribes J, Borras J. Epidemiology of primary liver cancer. *Semin Liver Dis.* 1999;19(3):271-285.

68. Simonetti RG, Camma C, Fiorello F, Politi F, D'Amico G, Pagliaro L. Hepatocellular carcinoma. A worldwide problem and the major risk factors. *Dig Dis Sci.* 1991;36(7):962-972.

69. Rich NE, Yopp AC, Singal AG, Murphy CC. Hepatocellular Carcinoma Incidence Is Decreasing Among Younger Adults in the United States. *Clin Gastroenterol Hepatol.* 2020;18(1):242-248 e245.

70. Chen W, Zheng R, Zhang S, et al. Cancer incidence and mortality in China in 2013: an analysis based on urbanization level. *Chin J Cancer Res.* 2017;29(1):1-10.

71. Venketasubramanian N, Yoon BW, Pandian J, Navarro JC. Stroke Epidemiology in South, East, and South-East Asia: A Review. *J Stroke.* 2017;19(3):286-294.

72. Siegel RL, Miller KD, Fuchs HE, Jemal A. Cancer statistics, 2022. *CA Cancer J Clin.* 2022;72(1):7-33.

73. Ayuso C, Rimola J, Vilana R, et al. Diagnosis and staging of hepatocellular carcinoma (HCC): current guidelines. *Eur J Radiol.* 2018;101:72-81.

74. Mourad A, Deuffic-Burban S, Ganne-Carrie N, et al. Hepatocellular carcinoma screening in patients with compensated hepatitis C virus (HCV)-related cirrhosis aware of their HCV status improves survival: a modeling approach. *Hepatology.* 2014;59(4):1471-1481.

75. Ayuso C, Rimola J, Garcia-Criado A. Imaging of HCC. *Abdom Imaging.* 2012;37(2):215-230.

76. Silva MA, Hegab B, Hyde C, Guo B, Buckels JA, Mirza DF. Needle track seeding following biopsy of liver lesions in the diagnosis of hepatocellular cancer: a systematic review and meta-analysis. *Gut.* 2008;57(11):1592-1596.

77. Marrero JA, Feng Z, Wang Y, et al. Alpha-fetoprotein, des-gamma carboxyprothrombin, and lectin-bound alpha-fetoprotein in early hepatocellular carcinoma. *Gastroenterology.* 2009;137(1):110-118.

78. Mitchell PS, Parkin RK, Kroh EM, et al. Circulating microRNAs as stable blood-based markers for cancer detection. *Proc Natl Acad Sci U S A.* 2008;105(30):10513-10518.

79. Taketa K. Alpha-fetoprotein: reevaluation in hepatology. *Hepatology.* 1990;12(6):1420-1432.

80. Chen VL, Sharma P. Role of Biomarkers and Biopsy in Hepatocellular Carcinoma. *Clin Liver Dis.* 2020;24(4):577-590.

81. Abelev GI, Perova SD, Khramkova NI, Postnikova ZA, Irlin IS. Production of embryonal alpha-globulin by transplantable mouse hepatomas. *Transplantation.* 1963;1:174-180.

82. IuS T. [Detection of Embryo-Specific Alpha-Globulin in the Blood Serum of a Patient with Primary Liver Cancer]. *Vopr Med Khim.* 1964;10:90-91.

83. Masopust J, Kithier K, Radl J, Koutecky J, Kotal L. Occurrence of fetoprotein in patients with neoplasms and non-neoplastic diseases. *Int J Cancer.* 1968;3(3):364-373.

84. Zheng Y, Zhu M, Li M. Effects of alpha-fetoprotein on the occurrence and progression of hepatocellular carcinoma. *J Cancer Res Clin Oncol.* 2020;146(10):2439-2446.

85. Bai DS, Zhang C, Chen P, Jin SJ, Jiang GQ. The prognostic correlation of AFP level at diagnosis with pathological grade, progression, and survival of patients with hepatocellular carcinoma. *Sci Rep.* 2017;7(1):12870.

86. Tellapuri S, Sutphin PD, Beg MS, Singal AG, Kalva SP. Staging systems of hepatocellular carcinoma: A review. *Indian J Gastroenterol.* 2018;37(6):481-491.

87. Llovet JM, Bru C, Bruix J. Prognosis of hepatocellular carcinoma: the BCLC staging classification. *Semin Liver Dis.* 1999;19(3):329-338.

88. Hanahan D, Weinberg RA. The hallmarks of cancer. *Cell.* 2000;100(1):57-70.

89. Fidler IJ. The pathogenesis of cancer metastasis: the 'seed and soil' hypothesis revisited. *Nat Rev Cancer.* 2003;3(6):453-458.

90. Etzioni R, Penson DF, Legler JM, et al. Overdiagnosis due to prostate-specific antigen screening: lessons from U.S. prostate cancer incidence trends. *J Natl Cancer Inst.* 2002;94(13):981-990.

91. Dong F, Budhu AS, Wang XW. Translating the metastasis paradigm from scientific theory to clinical oncology. *Clin Cancer Res.* 2009;15(8):2588-2593.

92. Jones S, Chen WD, Parmigiani G, et al. Comparative lesion sequencing provides insights into tumor evolution. *Proc Natl Acad Sci U S A.* 2008;105(11):4283-4288.

93. Poon RT, Fan ST, Ng IO, Lo CM, Liu CL, Wong J. Different risk factors and prognosis for early and late intrahepatic recurrence after resection of hepatocellular carcinoma. *Cancer.* 2000;89(3):500-507.

94. Izumi R, Shimizu K, Ii T, et al. Prognostic factors of hepatocellular carcinoma in patients undergoing hepatic resection. *Gastroenterology.* 1994;106(3):720-727.

95. Forner A, Llovet JM, Bruix J. Hepatocellular carcinoma. *Lancet.* 2012;379(9822):1245-1255.

96. Kurokawa Y, Matoba R, Takemasa I, et al. Molecular-based prediction of early recurrence in hepatocellular carcinoma. *J Hepatol.* 2004;41(2):284-291.

97. Tabrizian P, Jibara G, Shrager B, Schwartz M, Roayaie S. Recurrence of hepatocellular cancer after resection: patterns, treatments, and prognosis. *Ann Surg.* 2015;261(5):947-955.

98. Imamura H, Matsuyama Y, Tanaka E, et al. Risk factors contributing to early and late phase intrahepatic recurrence of hepatocellular carcinoma after hepatectomy. *J Hepatol.* 2003;38(2):200-207.

99. Hoshida Y, Villanueva A, Kobayashi M, et al. Gene expression in fixed tissues and outcome in hepatocellular carcinoma. *N Engl J Med.* 2008;359(19):1995-2004.

100. Ikai I, Arii S, Kojiro M, et al. Reevaluation of prognostic factors for survival after liver resection in patients with hepatocellular carcinoma in a Japanese nationwide survey. *Cancer.* 2004;101(4):796-802.

101. Poon RT, Fan ST, Lo CM, et al. Extended hepatic resection for hepatocellular carcinoma in patients with cirrhosis: is it justified? *Ann Surg.* 2002;236(5):602-611.

102. Ishizawa T, Hasegawa K, Aoki T, et al. Neither multiple tumors nor portal hypertension are surgical contraindications for hepatocellular carcinoma. *Gastroenterology.* 2008;134(7):1908-1916.

103. Roayaie S, Obeidat K, Sposito C, et al. Resection of hepatocellular cancer </=2 cm: results from two Western centers. *Hepatology.* 2013;57(4):1426-1435.

104. Minagawa M, Makuuchi M, Takayama T, Kokudo N. Selection criteria for repeat hepatectomy in patients with recurrent hepatocellular carcinoma. *Ann Surg.* 2003;238(5):703-710.

105. Golub TR, Slonim DK, Tamayo P, et al. Molecular classification of cancer: class discovery and class prediction by gene expression monitoring. *Science.* 1999;286(5439):531-537.

106. Mostertz W, Stevenson M, Acharya C, et al. Age- and sex-specific genomic profiles in non-small cell lung cancer. *JAMA.* 2010;303(6):535-543.

107. Mathe EA, Nguyen GH, Bowman ED, et al. MicroRNA expression in squamous cell carcinoma and adenocarcinoma of the esophagus: associations with survival. *Clin Cancer Res.* 2009;15(19):6192-6200.

108. Hoshida Y, Nijman SM, Kobayashi M, et al. Integrative transcriptome analysis reveals common molecular subclasses of human hepatocellular carcinoma. *Cancer Res.* 2009;69(18):7385-7392.

109. Scherf U, Ross DT, Waltham M, et al. A gene expression database for the molecular pharmacology of cancer. *Nat Genet.* 2000;24(3):236-244.

110. Thorgeirsson SS, Grisham JW. Molecular pathogenesis of human hepatocellular carcinoma. *Nat Genet.* 2002;31(4):339-346.

111. Marquardt JU, Andersen JB, Thorgeirsson SS. Functional and genetic deconstruction of the cellular origin in liver cancer. *Nat Rev Cancer.* 2015;15(11):653-667.
112. Libbrecht MW, Noble WS. Machine learning applications in genetics and genomics. *Nat Rev Genet.* 2015;16(6):321-332.
113. Sara H, Kallioniemi O, Nees M. A decade of cancer gene profiling: from molecular portraits to molecular function. *Methods Mol Biol.* 2010;576:61-87.
114. Italiano A. Prognostic or predictive? It's time to get back to definitions! *J Clin Oncol.* 2011;29(35):4718; author reply 4718-4719.
115. Villanueva A, Hoshida Y, Battiston C, et al. Combining clinical, pathology, and gene expression data to predict recurrence of hepatocellular carcinoma. *Gastroenterology.* 2011;140(5):1501-1512 e1502.
116. Lee JS, Heo J, Libbrecht L, et al. A novel prognostic subtype of human hepatocellular carcinoma derived from hepatic progenitor cells. *Nat Med.* 2006;12(4):410-416.
117. Segal E, Friedman N, Koller D, Regev A. A module map showing conditional activity of expression modules in cancer. *Nat Genet.* 2004;36(10):1090-1098.
118. Naugler WE, Sakurai T, Kim S, et al. Gender disparity in liver cancer due to sex differences in MyD88-dependent IL-6 production. *Science.* 2007;317(5834):121-124.
119. In: Hoshida Y, ed. *Hepatocellular Carcinoma: Translational Precision Medicine Approaches.* Cham (CH)2019.
120. Yeu Y, Yoon Y, Park S. Protein localization vector propagation: a method for improving the accuracy of drug repositioning. *Mol Biosyst.* 2015;11(7):2096-2102.
121. Xue H, Li J, Xie H, Wang Y. Review of Drug Repositioning Approaches and Resources. *Int J Biol Sci.* 2018;14(10):1232-1244.
122. Ashburn TT, Thor KB. Drug repositioning: identifying and developing new uses for existing drugs. *Nat Rev Drug Discov.* 2004;3(8):673-683.
123. Wilhelm SM, Carter C, Tang L, et al. BAY 43-9006 exhibits broad spectrum oral antitumor activity and targets the RAF/MEK/ERK pathway and receptor tyrosine kinases involved in tumor progression and angiogenesis. *Cancer Res.* 2004;64(19):7099-7109.
124. Bruix J, Takayama T, Mazzaferro V, et al. Adjuvant sorafenib for hepatocellular carcinoma after resection or ablation (STORM): a phase 3, randomised, double-blind, placebo-controlled trial. *Lancet Oncol.* 2015;16(13):1344-1354.

125. Minagawa M, Ikai I, Matsuyama Y, Yamaoka Y, Makuuchi M. Staging of hepatocellular carcinoma: assessment of the Japanese TNM and AJCC/UICC TNM systems in a cohort of 13,772 patients in Japan. *Ann Surg.* 2007;245(6):909-922.

126. Parvathaneni V, Kulkarni NS, Muth A, Gupta V. Drug repurposing: a promising tool to accelerate the drug discovery process. *Drug Discov Today.* 2019;24(10):2076-2085.

127. Oprea TI, Tropsha A, Faulon JL, Rintoul MD. Systems chemical biology. *Nat Chem Biol.* 2007;3(8):447-450.

128. Gupta SC, Sung B, Prasad S, Webb LJ, Aggarwal BB. Cancer drug discovery by repurposing: teaching new tricks to old dogs. *Trends Pharmacol Sci.* 2013;34(9):508-517.

129. Boyer A, Pasquier E, Tomasini P, et al. Drug repurposing in malignant pleural mesothelioma: a breath of fresh air? *Eur Respir Rev.* 2018;27(147).

130. Tang G, Cho M, Wang X. OncoDB: an interactive online database for analysis of gene expression and viral infection in cancer. *Nucleic Acids Res.* 2022;50(D1):D1334-D1339.